

Program overview

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Year 2022/2023
 Organization Architecture and the Built Environment
 Education Master Geomatics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Geomatics							
MSc 1 GM							
GEO1000	Python Programming for Geomatics	5					
GEO1001	Sensing Technologies	5					
GEO1002	Geographical Information Systems (GIS) and Cartography	5					
GEO1003	Positioning and Location Awareness	5					
GEO1006	Geo Database Management Systems	5					
GEO1015	Digital Terrain Modelling	5					
Starting Course MSc1							
AR071	Workshops Faculty of Architecture and the Built Environment	0					
MSc 2 GM							
GEO1004	3D Modelling of the Built Environment	5					
GEO1007	Geoweb Technology	5					
GEO1009	Geo-information Governance	5					
GEO1016	Photogrammetry and 3D Computer Vision	5					
MSc 2 GM - Electives 10 EC							
MSc 3 GM							
GEO1101	Synthesis Project	10					
GEO2011	Thesis Preparation	10					
MSc 3 GM - Electives 10 EC							
MSc 4 GM							
GEO2020	Thesis	30					
GM Electives							
GM Electives, courses listed below are recommended to be chosen as electives							
GEO5010	Research Assignment	2					
GEO5012	Land Administration	5					
GEO5014	Geomatics as support for energy applications	5					
GEO5015	Modelling wind and dispersion in urban environments	5					
GEO5016	Geomatics in practice	10					
GEO5017	Machine Learning for the Built Environment	5					
GEO5018	Ethics for the Data-driven City	5					
Electives, courses listed below are recommended to be chosen as electives							
AP3421	Fundamentals of Quantum Information	4					
AR0771	Beyond 3D Computer Visualisation	6					
CESE4055	Ad-hoc Networks	5					
CESE4065	Advanced Practical I.o.T. and Seminar	5					
CESE4075	Supercomputing for Big Data	5					
CIE4510-20	Climate Change: Science and Ethics	4					
CIE4522-15	GPS for Civil Engineering and Geosciences	4					
CIE4601	Physics of the Earth and Atmosphere	5					
CIE4602	Cryosphere: Remote Sensing and Modelling	4					
CIE4603-16	Geo-signal Analysis	6					
CIE4606	Geodesy and Remote Sensing	5					
CIE4607	Ocean topography and sea-level change	4					
CIE4609	Geodesy and natural hazards	4					
CIE4611	Geo-measurement processing	5					
CIE4615	Geoscience and Remote Sensing fieldwork	3					
CIE4616	Remote Sensing and Big Data	5					
CIE5401	GIS & Remote Sensing for Water Resources	3					
CIE5817	Assessment of Transport Infrastructure and Systems	4					
CS4375	Artificial Intelligence Techniques	5					
EE4740	Data Compression: Entropy and Sparsity Perspectives	5					
ET4169	Radar I: From Basic Principles to Applications	5					
IN4252	Web Science & Engineering	5					
IN4302TU	Building Serious Games	5					
IN4325	Information Retrieval	5					
RO47003	Robot Software Practicals	5					
SEN9235	Game Design Project	5					
WI4227-14	Discrete Optimisation	6					

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MSc 1 GM

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Instructor	Dr. D. Peng	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able: 1. to explain and use the basic elements of a programming language; 2. to describe and give examples of some Object Oriented programming features; 3. to translate a (simple) problem into an algorithm; 4. to construct a correctly functioning program; 5. to understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). Both assignments and exam have to be graded sufficient to pass the course.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO1001	Sensing Technologies	5
Course Coordinator	Dr. A. Rafiee	
Instructor	Ir. E. Verbree	
Responsible for assignments	Dr. A. Rafiee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra & Calculus at the level of the TUDelft courses Lineaire Algebra 1 and Analyse 1; basics of statistics.	
Course Contents	This course provides an overview on the principles and applications of spaceborne, airborne and terrestrial sensing technologies for geographic data acquisition as well as techniques for processing and information extraction from the acquired data. The topics of this course include: Physics of Remote Sensing: Electromagnetic(EM) spectrum and EM interaction with atmosphere and earth surface Satellite imagery data processing and machine learning techniques for thematic information extraction Principles and applications of RADAR and SONAR Principles of Airborne and terrestrial Lidar, introduction to point cloud data acquisition techniques and their applications in different domains Principles of Photogrammetry (acquisition, aerotriangulation, adjustment, Stereophotogrammetry, orthophoto and quality assessment) Introduction to positioning systems Introduction on preparing sensor outputs for GIS storage: (1) image matching techniques; (2) digital image processing; (3) machine learning for automated information extraction and quality assessment approaches including (4) least squares adjustment (LSA); (5) surface fitting for outlier detection.	
Study Goals	After the course the student is able to Describe different sensing techniques, from different platforms, for acquiring geographic data (what is where and when) Explain the applied metrics for quality assessment of the acquired data Compare different sensing techniques for their suitability in different application domains from their efficiency and quality points of view Apply data processing and machine learning techniques on the (self) acquired geographic data Analyse the applied data processing/ information extraction algorithms and evaluate their capabilities/drawbacks Select sensing technologies for a decision making problem	
Education Method	Lectures 30 hours; 26 hours Assignments; 84 hours self-study. The assignments aim to support gaining insight into the diverse techniques, methods and concepts explained during the lectures. Students will also participate in a 2-days excursion to Intergeo, the biggest geomatics event worldwide. In 2022 the event will be held on 18-20 October in Essen, Germany (hybrid).	
Books	Handouts and open books will be provided to the students	
Assessment	Written exam and Assignments. The result of the written exam determines 70% of the final score and the Assignments 30%.	
Period of Education	Quarter	
Leerstoel	GIS Technologie	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	The overall objectives of the course are:	
	a) To understand what a GIS is (e.g. listing its different parts and explaining what can be done with it); b) To identify, classify and evaluate the various internal running processes and algorithms used when, for example, a user "clicks on a button" in a GIS package; c) To apply GIS knowledge to solve practical spatial problems with it.	
	After the course the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geodatasets	
	3) Describe and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	4) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	5) List the main spatial data structures used in GIS, compare and discuss them;	
	6) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	7) Given a specific geographical problem, identify and analyse which GIS approach (in terms of data structure and algorithms) is the most suitable, and justify the reason for choosing it;	
	8) Generalise the GIS knowledge to solve more complex spatial problems by integrating the existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package).	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO1003	Positioning and Location Awareness	5
Course Coordinator	Ir. E. Verbree	
Instructor	Dr. A. Rafiee	
Responsible for assignments	Ir. E. Verbree	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course addresses Global Navigation Satellite Systems (GNSS) and (indoor) positioning technologies for sensing people, devices, and assets in the built environment with the focus on location-aware applications. The course covers the requirements and context for these location-aware applications: global, local, and linear reference systems, coordinate systems and map projections, positioning methods and techniques, and the social and technical push and legislative pull factors that empower the development of location-based services. The course consists of three parts: 1. Analyses of location aware applications: concepts of location awareness, location sensitivity, and context awareness; use cases of location-based applications; technical and legislation factors, privacy issues, and standards. 2. Global, local (e.g. Dutch), and linear reference systems; coordinate systems and transformations; basic types of map projections. 3. Positioning methods, techniques, and performance: Global Navigation Satellite Systems; terrestrial radio-based positioning by using Bluetooth, UWB, RFID, Wi-Fi and other short range technologies; Inertial Navigation Systems (INS); presence-based locating; technologies with dead reckoning and map-matching methods; fingerprinting (templates) for indoor positioning.	
Study Goals	After the course Positioning and Location Awareness the student is able to: 1. Understand location awareness, location sensitivity, context awareness; 2. Understand the different types of reference systems: global, local (Dutch), linear; 3. Understand the ethical and legislative factors of methods for location awareness; 4. Apply different coordinate systems, positioning, and indoor localisation; 5. Evaluate different technologies to support location awareness on their technical performance (availability, accuracy, integrity, continuity), and their ethical factors and legislative factors (privacy issues).	
Education Method	Lectures: 28 hours Assignments: 72 hours Self-study: 40 hours	
Assessment	- Group Assignments (20%) - Individual Assignments (20%) - Written Exam (60%) Students need to get at least a sufficient grade for both the exam and the assignment in order to obtain the overall grade for the course.	
Period of Education	Quarter	
Used Materials	Reader	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO1006	Geo Database Management Systems	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	Drs. C.W. Quak	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Contact Hours / Week x/x/x/x	2*2 hours lectures and 3 hour (assignment support + feedback)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Wiskunde B VWO (Mathematics B of Dutch VWO education) Time/space complexity of algorithms Binary number representation Operating system command line (e.g. of Windows, Linux) Geomatics courses: GEO1002 (GIS and Cartography) and GEO1000 (Python Programming)	
Course Contents	This course is about managing geo-information in a database management system (DBMS). The geo-DBMS is the central component of the geo-information chain. All information observed and interpreted should be managed in a manner that supports further use, so the Geo-DBMS is the end-point of data collected. Analysis, visualizing and (web-) dissemination of the geo-information start from the geo-DBMS. It is crucial that the content of the geo-DBMS is clear (the information model and its associated semantics) to the users, but also that it is high performant, as the geo-data volumes are often very high and especially in a web-based environment there may be many uses. The course consists of three parts; the first two concern introductions. The main part is devoted to spatial modelling and geo-DBMS. More specifically the three parts cover: 1. Introduction object-oriented information modelling: UML and more specifically class diagram to create conceptual models. Mastering (and applying in design) the key notions such as: Class, Abstract Class, Identifiers, Instances, Attribute, Operation, Association, Generalization, Aggregation, Composition, Multiplicity, Note, Constraint, Package, Stereotype, etc. 2. Introduction relational database management systems: the relational model, SQL DDL (schema definition) and DML (queries). Relational algebra. Some implementation issues (clustering and indexing techniques, such as the b-tree, primary and foreign key constructions supporting referential integrity). In addition a first introduction to NoSQL (Not Only SQL) databases. 3. Spatial databases: theoretical aspects of spatial modelling (object and field models), DBMS-GIS architectures (dual, layered, integrated), spatial-temporal data model, Abstract Data Types (ADTs) for geometric primitives, topology model (ISO 19107), temporal aspects, spatial indexing (quad tree, r-tree, field-tree), spatial data clustering (Hilbert, Morton, Cantor, row, row prime, etc.), geo-DBMSs (data modelling, structures, queries, visualisation), (non-)commercial systems (Oracle Spatial, PostGIS, etc.). One or more advanced topics: 3D, Raster, Point cloud data, Simplicial complexes, Geo-OCL, Big data, NoSQL (spatial), Vario-scale, 5D modelling (deeply integrated 3D space, time and scale).	
Study Goals	After the course the student is able to: - Students can infer the implications for the real world situation that is described by the diagram complex UML class diagrams. - Students can convert a descriptive text into a UML diagrams - Students can create a DBMS schema to store the information for a real life situation. - Students can apply the SQL Query language to query a DBMS: More specific: A student knows how to: join two or more tables, How to aggregate data, make meaningful selections, aggregate data, use subqueries. - Students can transform simple SQL statements into relations algebra that contain the following algebraic operations: SELECT PROJECT JOIN UNION DIFFERENCE - Students knows the difference between: primary keys, foreign keys, indexing and clustering and knows to when apply these techniques to improve database performance for a given database. - Students can program functionality inside a DBMS with a procedural language (PL/pgSQL in PostgreSQL) - Students have some direct experiences with UML modelling tool (Enterprise Architect), SQL database (PostgreSQL) and have seen some NoSQL database examples (Neo4j, MongoDB). - Students have some direct experiences with retrieval (select) and update information from the database using SQL (including joins) and optimise the implementation based on standard database techniques (primary/foreign key, indexing and clustering) - Students understand the differences of spatial data and design a conceptual model for a spatial information system, translate the spatial conceptual model in a relational database with spatial extension, optimize the implementation of spatial databases (spatial clustering indexing, spatial constraints,...) and retrieve and update spatial data using spatial operations (both geometric and topological) in the selection in combination with non-spatial predicates - Students understand some advanced topics: spatial-temporal modelling, 3D modelling, routing inside the DBMS (using pgRouting) vario- or multiscale modelling, Spatial OCL (Object Constraint Language) formalization, routing, TIN, point clouds, efficient raster data management, and 5D modelling.	
Education Method	Lectures per week: 4 hours theory and 3 hour (assignment support + feedback). Lectures: 28 hours Assignments (supervised): 21 hours Self-study: 91 hours (includes reading and practical assignments, unsupervised parts) The assignments require the use of software: Enterprise Architect, PostgreSQL/PostGIS and Quantum GIS.	
Literature and Study Materials	- Textbook (Elmasri & Navathe, Fundamentals of Database systems, 5th. Int. Edition, Addison-Wesley, ISBN: 0-321-41506-X). - PostgreSQL user manual - PostGIS manual - Randy Miller, Practical UML: A Hands-On Introduction for Developers. - Scott W. Ambler, UML Class Diagrams, excerpt from: The Object Primer, 3rd Edition: Agile Model Driven Development with UML 2 - Tom Vrijlbrief and Peter van Oosterom (1992). GEO++: An extensible GIS. In proceedings 5th International Symposium on Spatial Data Handling, Charleston, South Carolina, pages 40-50, August 1992. - Section 6.1 (Geometry object model, esp. 6.1.15 Relational operators) and section 7.2.6 (Examples) of OGC SFS (2011), that is, the OpenGIS® Implementation Standard for Geographic information - Simple feature access - Part 1: Common architecture. - Eliseo Clementini, Paolino Di Felice and Peter van Oosterom. A Small Set of Formal Topological Relationships Suitable for End-User Interaction. The 3rd International Symposium on Large Spatial Databases, Singapore, (LNCS nr. 692), pages 277-295, June 1993. - Peter van Oosterom, Wilko Quak and Theo Tijssen (2004). About Invalid, Valid and Clean Polygons. In: Peter F. Fisher(Ed.); Developments in Spatial Data Handling, 11th International Symposium on Spatial Data Handling, 2004. - Peter van Oosterom (2006). Constraints in Spatial Data Models, in a Dynamic Context. In: J. Drummond, R. Billen, E. Joao and D. Forrest (Eds.); Dynamic and Mobile GIS: Investigating Changes in Space and Time, pp. 104-137 - Peter van Oosterom. Spatial Access Methods. Chapter T2.3 in Geographical Information Systems Principles, Technical Issues, Management Issues, and Applications (edited by Longley, Goodchild, Maguire en Rhind), Wiley pages 385-400 (vol.1), 1999. - Peter van Oosterom and Tom Vrijlbrief. 'The Spatial Location Code', presented at SDH'96, Delft 12-16 August 1996. - P.J.M. van Oosterom, B. Maessen and C.W. Quak (2002). Generic query tool for spatio-temporal data. In: International Journal of Geographical Information Science, Volume 16, 8, 2002, pp. 713-748 - Peter van Oosterom, Hendrik Ploeger, Jntien Stoter, Rod Thompson and Christiaan Lemmen (2006). Aspects of a 4D Cadastre: A First Exploration. XXIII International FIG congress, Munich, October 2006, 23 p. - Friso Penninga and Peter van Oosterom (2008). A simplicial complex-based DBMS approach to 3D topographic data modeling. In: International Journal of Geographical Information Science, Volume 22, 7, 2008, pp. 751-779. - Peter van Oosterom (2005). Variable-scale Topological Data Structures Suitable for Progressive Data Transfer: The GAP-face	

	Tree and GAP-edge Forest. In: Cartography and Geographic Information Science, Volume 32, 4, 2005, pp. 331-346 - Peter van Oosterom and Martijn Meijers (2014). Vario-scale data structures supporting smooth zoom and progressive transfer of 2D and 3D data. In: International Journal of Geographical Information Science, Volume 28(3), 455-478. - Lina Huang, Martijn Meijers, Radan uba, Peter van Oosterom (2016). Engineering web maps with gradual content zoom based on streaming vector data, In: ISPRS Journal of Photogrammetry and Remote Sensing, Elsevier BV, 114, pp. 274-293, 2016. - Peter van Oosterom, Oscar Martinez-Rubi, Milena Ivanova, Mike Horhammer, Daniel Geringer, Siva Ravada, Theo Tijssen, Martin Kodde, and Romulo Goncalves (2015). Massive point cloud data management: design, implementation and execution of a point cloud benchmark. In Computers&Graphics. - Peter van Oosterom, Oscar Martinez-Rubi, Theo Tijssen, Romulo Goncalves (2016). Realistic Benchmarks for Point Cloud Data Management Systems, Chapter in: Advances in 3D Geoinformation, Springer Nature, pp. 1-30.
Assessment	Written exam (3 hours) and lab assignments. The exam result will determine the final score, but lab assignments need to be sufficient. The main purpose of lab assignment is to support the understanding of the theory via hands-on experience.
Period of Education	Second Quarter
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/

GEO1015	Digital Terrain Modelling	5
Course Coordinator	H. Ledoux	
Instructor	G.A.K. Arroyo Ohoi	
Responsible for assignments	H. Ledoux	
Contact Hours / Week	4h/week	
x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Digital terrain models (DTMs) are computer representations of the elevation of a given area, and they play an important role in understanding and analysing our built environment. They are the necessary input for several applications (eg flood modelling, visibility, effects of climate change on the north poles, etc.), and they are also relevant for studying the seabed. The course provides an overview of the fundamentals of digital terrain modelling: - different representations of terrains: TINs, rasters, point clouds, contour lines - reconstruction of terrains from different sources (lidar, InSAR, photogrammetry, multibeam echosounders) - spatial interpolation methods - conversion between different representations - processing of terrains and point clouds: outlier detection, filtering, segmentation, and identification and classification of objects - techniques to handle and process massive datasets - applications, eg runoff modelling, watersheds computations, visibility The course has both a theoretical part and a practical part where students reconstruct, manipulate, process, and extract information from terrains. All the labs are programming tasks (to be done with Python or C++), based solely on open-source libraries and software.	
Study Goals	At the end of the course, students will be able to: - describe the characteristics of elevations datasets from different sources (LiDAR, photogrammetry, InSAR) - describe the pros and cons of different representations of terrains, and compare them for different applications - explain how elevation datasets can be automatically converted to terrains - reconstruct and manipulate terrains using with open-source libraries - explain, analyse, and discuss how terrains can be useful in different applications related to built environment - given a specific problem where elevation plays a role (eg visibility or flood modelling), analyse and identify which data and algorithms are needed to solve the problem, and assess the consequences of these choices	
Education Method	We use a "flipped classroom": each week there are 2 main topics, and students first watch the videos and read the material at home. Then during the contact hours (2 x 2h), the most difficult parts are discussed and students get help/support for the practicals.	
Books	The open book "Computational modelling of terrains", written by the instructors of the course, is used. PDF and LaTeX source code at: https://github.com/tudelft3d/terrainbook/	
Prerequisites	The course is designed for students from the MSc Geomatics, and the following courses are required prerequisites: (1) GEO1000 (or knowledge of scripting/programming in at least one language (eg Matlab, C++, or Python); (2) GEO1001; (3) GEO1002. Students from other faculties with the appropriate background are welcome.	
Assessment	- two written exams (1 mid-term (at week 2.5) + 1 final exam) - two or three assignments (programming in Python or C++)	
Remarks	All material can be found at https://3d.bk.tudelft.nl/courses/geo1015/ Brightspace will not be used.	
Period of Education	Q2	

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Organization	Architecture and the Built Environment
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Starting Course MSc1

AR071	Workshops Faculty of Architecture and the Built Environment	0
Course Coordinator	Dr.ir. M.J. Hoekstra	
Contact Hours / Week x/x/x/x	X / 0 / 0 / 0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	All first year Master students of the Faculty of Architecture and the Built Environment will start the academic year 2021-2022 with a MSc Kick Off programme on Friday 27 August and Saturday afternoon 28 August 2021. With a mix of lectures, workshops and sessions guided by teachers of the faculty, you will e.g. be introduced to (design) ethics, scientific integrity and/or intercultural communication. With this programme you will make a first start to cover the ethics engineering learning goals of the Master programmes. Further, we wish to enhance the interaction between all new students, both Dutch and International, and to introduce you to settings, methods and procedures of the faculty. Participation in the programme is highly recommended for all students starting their Master 1 programme in September.	
Study Goals	- The student has a basic understanding of moral sensibility, moral analysis skills, moral creativity, moral judgement skills, moral decision-making skills and moral argumentation skills.	
Education Method	Lectures, workshops, games.	
Assessment	Not applicable	
Special Information	Please note that this programme starts in the week before the Opening of the Academic year. The MSc Kick Off programme will be held on Friday 27 August and Saturday afternoon 28 August 2021. For more information see website: https://www.tudelft.nl/studenten/faculteiten/bk-studentenportal/onderwijs/master-of-science/master-kick-off/ As a consequence of the global covid-19 pandemic, we have downscaled the workshops and sessions in this course. For more detailed information, we refer you to the email we will be sending you in the 2nd half of August 2021 in regards to the introduction days.	
Period of Education	1,5 days	
Concept Schedule	Two days: Friday and Saturday before start academic year	

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

MSc 2 GM

GEO1004	3D Modelling of the Built Environment	5
Course Coordinator	G.A.K. Arroyo Ohoi	
Instructor	R.Y. Peters	
Instructor	H. Ledoux	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.7	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides a detailed description of the main ways in which the built environment is modelled in three dimensions, covering material from low-level data structures for generic 3D data to high-level semantic data models for cities. The main topics described in the course are: - the difference between reality, data models and data structures; - how geometry, topology and attributes can be stored in 3D models; - boundary representation (b-rep) and meshes; - (constrained) tetrahedralisation and 3D Voronoi diagrams; - voxel models and voxelisation; - constructive solid models (CSG) and Nef polyhedra; - Bézier surfaces; - the medial axis transform (MAT); - generalised (g-) and combinatorial maps (c-maps); - the ISO 19107 geometry representations; - semantic 3D city models and CityJSON; - building information modelling and IFC; - building reconstruction for 3D city models; - applications of 3D models (eg calculation of volumes and solar potential); The course has both a theoretical part based on short lectures, watching videos and reading the provided material, as well as a practical part where students create, manipulate and view 3D models, and then use them for different applications. All the labs are programming tasks (to be done with the C++ programming language), based solely on open-source libraries and software.	
Study Goals	At the end of the course, students should be able to: 1. explain the difference between semantic data models, data structures and reality, and implement solutions that combine these levels; 2. compare different modelling approaches, such as boundary representation and volumetric representations, outline their relative merits and drawbacks, and choose an appropriate approach for a given use case; 3. interpret topological properties like 2-manifoldness, and execute solutions that use these properties to store 3D models; 4. implement several different data structures for the storage of 3D models; 5. outline the characteristics of the main semantic open data models used in GIS (CityGML-CityJSON) and BIM (IFC), and to manipulate such models at a low level; 6. execute analyses (eg solar potential or volume calculations) based on a 3D city model and check their result.	
Education Method	Blended-learning: each week there are 2 main topics, and students first watch the videos and read the provided study material. Then, during the contact hours, the most difficult parts are discussed and students get help/support for the assignments.	
Literature and Study Materials	The course follows the 3D book, which is specifically developed for this course: https://github.com/tudelft3d/3dbook	
Prerequisites	The course is mainly designed for students from the MSc Geomatics, but everyone is welcome to follow the course. However, basic knowledge of GIS is recommended, and some knowledge of scripting/programming (in any language) is required to do the assignments.	
Assessment	- Two written examinations: 1 midterm + 1 final exam - C++ programming assignments (mix of analytical assignments and practical exercises) A weighted average of 50% in both exams and a total mark of 60% is necessary to pass the course.	
Remarks	All material can be found at https://3d.bk.tudelft.nl/courses/geo1004/ Brightspace will not be used.	
Period of Education	Quarter 3	

GEO1007	Geoweb Technology	5
Course Coordinator	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Wiskunde B VWO (Mathematics B of Dutch VWO education) Familiarity with operating system command line (e.g. of Windows, Mac, Linux) Geomatics courses: GEO1000 (Python Programming), GEO1001 (Sensing Technologies), GEO1002 (GIS and Cartography), GEO1003 (Location Awareness), GEO1006 (Geo-DBMS) and GEO1009 (Geo-Info Law)	
Course Contents	Today professionals and laymen alike have access to geo-data through the internet. Using a smart phone or tablet connected to broadband internet one can virtually fly-through a city and view buildings, underground constructions and proposed designs from any viewing angle, even augmented reality. The internet also strengthens the communication between government and citizens, who can comment on the designs, fill in questionnaires and vote, and in this way (local) government may increase public participation in city planning and other public affairs. Access to geo-web services should be fast & reliable and based on open standards such as OGC (Open Geospatial Consortium), ISO (International Organization for Standardization) and INSPIRE (European Spatial Data Infrastructure). The map interfaces should be user friendly while sufficient functionality should be available for carrying out analyses on the diverse datasets. A key tool when building geo-web services is Geography Markup Language (GML) an exchange language for geo-data based on XML (eXtensible Markup Language), one of the standard languages of the internet. The course consists of four parts: 1. Principles, application, evaluation and integration of generic web services and the importance of using standards when building them. The standards involved start with general ICT standards like HTTP, XML, JSON and SOAP (Simple Object Access Protocol). Increasingly the semantics of data in the framework of web services, and linked data, plays an important role. 2. Based on these general standards, geo standards and protocols can be build (ISO TC-211/OGC web services and protocols). Combining the geo standards and protocols results in a set of geo web services that more and more replace the traditional GIS tools. Topics of interest: geoweb system server-client architecture, WMS/WFS/ server, desktop/browser/mobile clients (incl app development), web-based transaction processing, portrayal, query (filter encoding), tiling, metadata service, web processing service, vario-scale data services. 3. Principles and applications of (real-time) Sensor Web. The components of Sensor Web will be explained and (some of them) used: Observations & Measurements, Transducer Markup Language, Sensor Observation Service, Sensor Planning Service, Sensor Alert Service and the Web Notification Service. 4. Applying software tools for the visualization of 3D geo-data. Normally the first, and in many cases the most important, thing to do with (3D) geo-data is to visualize it. Nowadays data is delivered over the internet by means of web services. The standards, architecture and services for 3D geo-visualization, and the practical aspects of available (web) tools and how to use them will be explored.	
Study Goals	After the course the student is able to: - understand generic internet and web standards and apply these in the implementation of web information systems; - understand formal geo standards and derived implementation specifications / describe the existing geo web services and their application domains; - integrate several geo web services for supporting spatial decision making; - evaluate existing geo web services in terms of specified applications and give motivated suggestions for improvement; - understand the Sensor Web Enablement Common Data Framework (common data models and schema) and the SensorML (models and schema for sensor systems and processes surrounding measurements) and use them in web services; - describe the available (visualization) standards for the visualization of 3D geo-data, like X3D, 3D-pdf, WebGL, CityGML, KML, GeoJSON, glTF, b3dm, i3s and other industry standards / software tools.	
Education Method	Lectures: 28 hours Self-study: 48 hours Lab assignments (64 hours): on the various internet, web, geoweb standards, protocols, services and tools	
Literature and Study Materials	Selected chapters, standards documents & scientific and professional articles made available via Brightspace.	
Assessment	Written exam (3 hours), and lab assignments. The exam result will determine 60% of the final score, the lab assignments make up the other 40%.	
Period of Education	Fourth Quarter	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO1009	Geo-information Governance	5
Course Coordinator	Dr.ir. B. van Loenen	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course students will learn about the organisational and legal aspects relevant for developing a strategy for a geographic information infrastructure. Many organisations, and increasingly also citizens, collect, process and disseminate geo-information (e.g., Google, TomTom, mapping agencies and cadastres). Some information serves a specific purpose, while other information acts as multipurpose or base information, which is the fundament of an information infrastructure. An adequate information infrastructure allows for information to be collected efficiently (collect it once, use it many times) and provides reliable information for effective use in decision-making processes at all levels in government, private sector, academia and among citizens. Legislation is a critical component that needs to be considered to determine whether information can be used for a certain purpose and shared with others. The course consists of three parts: 1. Theoretical basis for the introduction, organisation and management of (geo-) information infrastructures. Aspects that will be addressed are stakeholders, co-ordination mechanisms and components of an information infrastructure, the affiliation with policy lines like electronic and efficient government. 2. Overview of legislation pivotal to developing geographic information infrastructures. Legal aspects applying to geo-information, such as intellectual property rights, privacy and access to government information legislation, constitute a major part of this course. 3. Practical application of the theory. Students are tasked to write a paper on a selected SDI topic.	
Study Goals	After this course the student is able to: - recognize and anticipate upon relevant legal (especially data protection (EU General Data Protection Regulation), copyright/database rights, access to and reuse of information law (Open Data and Reuse of Public Sector Information Directive), and data sharing legislation (INSPIRE Directive and Regulations)), and organisational issues evolving around the acquisition, processing, dissemination and use of (geographic) information - understand the needs of potential users of geographic information and their requirements within organisations - apply the concepts, processes and main components of (geographic) information infrastructures (Governance, Data, People, Policies, Access networks and Standards) to support geographic information sharing between organisations - critically assess geographic information management strategies for organisations - assess the performance of an (geographic) information infrastructure from a user perspective - (co-)author a scientific paper on a selected SDI topic - analyze moral problems in terms of facts, values, stakeholders and their interests - morally justify ones actions and to discuss and evaluate them together	
Education Method	Interactive lectures: 28 hours; Individual assignment: 42 hours; self-study: 70 hours.	
Assessment	Oral exam (max. 30 minutes); Group assignment (report and presentation). Report is 15 pages. The exam will determine 50% of the final score and the individual assignment 50%. Students need to get at least a sufficient grade (5,8) for both the exam and the group assignment in order to obtain the overall grade for the course. Group assignment consists of a mark for the paper and an individual reflection. The mark on the paper can be influenced positively/negatively for 0,5 grade point (maximum) by the individual reflection on the paper. Specific for the individual assignment part: Missing the deadline for the final paper results in a grade of 1,0 for the corresponding part/ the whole. Students need to present preliminary results of their research, need to review the paper of another student (group), and take part in the final debate in order to obtain a final grade for the overall paper assignment.	
Period of Education	Quarter 3	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO1016	Photogrammetry and 3D Computer Vision	5
Course Coordinator	Dr. L. Nan	
Contact Hours / Week x/x/x/x	6 hours/week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Photogrammetry and 3D computer vision aim at recovering the structure of real-world objects/scenes from images. This course is about the theories, methodologies, and techniques of 3D computer vision for the built environment. In the term of this course, students will learn the basic knowledge and algorithms in 3D computer vision through a series of lectures, reading materials, lab exercises, and group assignments. The topics cover the whole pipeline of reconstructing 3D models from images: - Cameras models: how a point from the real world gets projected onto the image plane and how to recover the camera parameters from a set of observations; - Epipolar geometry: the geometric relations between 3D points and their images points; the constraints between the image points; - Image matching: define and match image features (SIFT) to establish correspondences between images; - Structure from motion: recover/refine geometry and camera parameters from a set of images; - Multi-view stereo and learning-based approaches for recovering dense geometry (e.g., point clouds) from images; - Surface reconstruction: obtain 3D surface models of real-world objects from point clouds.	
Study Goals	After finishing this course, the students will have gained knowledge of 3D computer vision techniques and the skills to apply them for recovering 3D geometry of real-world objects from images. Specifically, the students will be able to: - apply linear algebra knowledge to implement basic 3D computer vision algorithms; - explain the main concepts in 3D computer vision (e.g., camera models, epipolar constraints, fundamental matrix, image matching, triangulation, structure from motion, bundle adjustment, multi-view stereo, and surface reconstruction); - explain the principles of the state-of-the-art 3D computer vision pipelines for 3D dense reconstruction from images; - choose appropriate methods for reconstructing smooth surfaces and piecewise planar objects; explain and evaluate these methods; - propose and implement solutions for reconstructing real-world buildings from images.	
Education Method	Lectures, reading materials, group assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is related to all fields of Geomatics for the Built Environment, from data acquisition, data management, data visualization, data analysis, 3D modeling, to applications.	
Literature and Study Materials	In addition to the lectures and handouts (distributed after each lecture), you will find relevant topics discussed in great detail in the following materials. [1] Y. Ma, et al. An Invitation to 3-D Vision: From Images to Models. 2001 [2] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Second Edition. Cambridge University Press, 2004. [3] D. Forsyth and J. Ponce. Computer Vision: A Modern Approach. Second Edition. Pearson, 2012. [4] R. Hartley and P. Sturm. Triangulation. Computer vision and image understanding. 1997. [5] Berthold K.P. Horn. Recovering baseline and orientation from essential matrix. 1990. [6] Y. Furukawa and J. Ponce. Accurate, Dense, and Robust Multiview Stereopsis. PAMI, 2010 [7] H. Hoppe, et al. Surface reconstruction from unorganized points. SIGGRAPH, 1992 [8] M. Kazhdan, M. Bolitho, H. Hoppe. Poisson surface reconstruction. SPG, 2006, 61-70 [9] L. Nan and P. Wonka. PolyFit: Polygonal Surface Reconstruction from Point Clouds. ICCV 2017.	
Practical Guide	In the assignments and group work sessions, students will experiment with the 3D computer vision techniques introduced in the lectures. - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; - Discuss the effect of parameters (if possible); - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what; - Strict deadline: 10% deduction per day late, no more accepted after 3 days.	
Prerequisites	- Basic linear algebra and calculus; - Basic knowledge and skills in C++ programming. Students who have no experience in the above topics can still participate but should be willing to make efforts and invest some more time.	
Assessment	The assessment of this course consists of three group assignments and the final written exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 0.5). - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Both assignments and the final exam have to meet the minimum requirement (i.e., 5.5) to pass the course. A total of 6.0 or above is necessary to pass the course.	
Period of Education	Quarter	
Course evaluation	See the "Educational quality assurance" at https://www.tudelft.nl/en/student/faculties/a-be-student-portal/education/educational-quality-assurance/	

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

MSc 2 GM - Electives 10 EC

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

MSc 3 GM

GEO1101	Synthesis Project	10
Course Coordinator	Dr.ir. B. van Loenen	
Instructor	Ir. E. Verbree	
Instructor	H. Ledoux	
Contact Hours / Week x/x/x/x	X/0/0/X	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	GEO2011/GEO2020	
Expected prior knowledge	Geomatics students should have gained 30 EC of core geomatics courses (Core courses are: GEO1000, GEO1001, GEO1002, GEO1003, GEO1004, GEO1006, GEO1007, GEO1009, GEO1015, and GEO1016). External students are admitted if they have similar knowledge and skills based on a certified course list.	
Course Contents	The project aims at investigating a real-world problem. Geomatics teachers in cooperation with a company, governmental agency, or a TU Delft research group that is not involved in the Geomatics program the clients define the problem. The problem will be addressed by a group of students. The project enables you to gain further knowledge and skills in acquisition, processing, analysis and interpretation, and visualization of geo-information as well as insights in the running of a project in which diverse stakeholders are involved.	
Study Goals	After this course students are able to: - Understand, through a project at hand, how each of the individual parts of the geo-information process (from measurement (data gathering), data processing and analysis, to presentation, delivery and application) relate to each other and to the chain as a whole. - Apply and expand the knowledge and competences acquired during the core Geomatics courses by developing a solution to a real world problem. - Apply the basic components of a project management strategy to a project addressing a real world problem. - Cooperate with students, tutors and professionals. - Evaluate the project result(s). - Present the project result(s) to peers and professionals in the field.	
Education Method	Group Project	
Course Relations	Geomatics Core Courses: GEO1000, GEO1001, GEO1002, GEO1003, GEO1004, GEO1006, GEO1007, GEO1009, GEO1015, and GEO1016.	
Reader	Project Guide	
Assessment	For this course students can obtain a pass or a fail. The decision (pass/fail) is determined in an individual meeting with the supervisors and based on the technical quality of the project, the individual reflection on the final report, the presentation on Geomatics day and the performance of the student in the research process. In order to pass the course, students also have to attend and take part in the group's midterm presentation and final presentation, and obtain a pass for their individual project strategy proposal. The pass/fail criteria are detailed in the rubrics available on Brightspace. If a student fails, then the student has to take the course in another academic year again.	
Enrolment / Application	BIS: only for Geomatics students (and Approved International 'Study Abroad' Students).	
Special Information	Deliverables are: - Project Strategy Proposal (individual task) - Project Identification Document (PID) (group task) - Oral mid-term presentation, including: (1) status and progress of the project; (2) proposed amendments to the PID if necessary (group task) - Final written report: A template will be provided to obtain uniformity in the structure of all synthesis reports (group task) - Reflection on the project including the final report (individual task) Oral reporting during Geomatics Day: - Presentation by the team including discussion (group task). You present your project to an extended audience of clients, tutors, all colleague students, and interested professionals.	
Period of Education	Quarter	
Concept Schedule	A kick-off meeting will be arranged at the end of Q4 (the quarter preceding the Q1 of the GEO1101 course concerned). This plenary session has to be attended by all students who want to carry out the synthesis project. The geomatics teachers who are your tutors will be present and, if possible, clients will be present/involved. The client or tutors will present the use case. You have two weeks to hand-over a weighted rank-order motivated list of the use-cases you prefer to participate in. In consultation with the tutors the coordinator will assign a project to you. In the first week of Q1 each project team and their tutors(s) and/ or the client will come together to discuss the use case.	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO2011	Thesis Preparation	10
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The graduation work is the last step in the Geomatics education. It is completed in the form of an individual research project. GEO2020 follows GEO2011, and involves carrying out the research according to the plan of GEO2011. The topic of the scientific project can be related to any of the subjects offered by the Geomatics programme. The project must be supervised by one or more of the lecturers who participated in the Geomatics education. The project can be performed in cooperation with a company/institution, which provides a use case or a problem statement (but the work is always supervised by the scientific staff of the university).	
Study Goals	The graduation work deliverables are: (1) a scientific report (a thesis); and (2) an oral presentation. With the graduation work the student had proven to comply with the generic and domain requirements of the Geomatics MSc degree such as: 1. has skilled in one or more academic disciplines; 2. has the ability to cooperate and communicate; 3. demonstrate scientific and technological skills; 4. show ability to work independently and perform research; 5. demonstrate capacity to manage, both technically and time-wise, a research or design project.	
Education Method	Individual work.	
Practical Guide	All details are available at: https://3d.bk.tudelft.nl/courses/geo2020/ Brightspace is not used.	
Assessment	scientific report + presentation. These are evaluated by the graduation committee, guided by the geomatics graduation manual.	
Special Information	Geomatics students have the possibility to carry out their graduation research project in collaboration with a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://3d.bk.tudelft.nl/courses/geo2020/company/. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	All.	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

MSc 3 GM - Electives 10 EC

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

MSc 4 GM

GEO2020	Thesis	30
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Contact Hours / Week x/x/x/x	X/X/X/X	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	GEO2011	
Course Contents	The graduation work is the last step in the Geomatics education. It is completed in the form of an individual research project. GEO2020 follows GEO2011, and involves carrying out the research according to the plan of GEO2011. The topic of the scientific project can be related to any of the subjects offered by the Geomatics programme. The project must be supervised by one or more of the lecturers who participated in the Geomatics education. The project can be performed in cooperation with a company/institution, which provides a use case or a problem statement (but the work is always supervised by the scientific staff of the university). The graduation work deliverables are: (1) a scientific report (a thesis); and (2) an oral presentation.	
Study Goals	With the graduation work the student had proven to comply with the generic and domain requirements of the Geomatics MSc degree such as: 1. has skilled in one or more academic disciplines; 2. has the ability to cooperate and communicate; 3. demonstrate scientific and technological skills; 4. show ability to work independently and perform research; 5. demonstrate capacity to manage, both technically and time-wise, a research or design project.	
Education Method	individual work	
Practical Guide	All details are available at: https://3d.bk.tudelft.nl/courses/geo2020/ Brightspace is not used.	
Assessment	scientific report + presentation. These are evaluated by the graduation committee, guided by the geomatics graduation manual.	
Special Information	Geomatics students have the possibility to carry out their graduation research project in collaboration with a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://3d.bk.tudelft.nl/courses/geo2020/company/. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	All	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

GM Electives

GEO5010	Research Assignment	2
Course Coordinator	C. Garcia Sanchez	
Instructor	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The course is meant as an introduction to one, or more, aspects of a research project. The student can do: 1. A literature study to get familiar with a related topic/domain that is not covered in the core programme of the MSc Geomatics. The knowledge learned can then be used as a start for a graduation project (GEO2010/2020), but it may also be unrelated to the thesis topic. 2. An implementation of concepts in a software prototype. This can be used for example to further explore one specific topic and to try to implement a prototype that will, for instance, allows us to process some datasets. This can be used to develop one's skills in a particular programming language. 3. A mix of 1 and 2 above. If a literature review is carried out, the main deliverable is a scientific report summarising the problem studied, the main findings, and potentially a discussion about future work related to the topic. If an implementation is carried out, then the student will deliver a software prototype, along with instructions about what it does, how to use it, test datasets, etc. Important note: If the research performed through GEO5010 is related to the MSc thesis, the work and output needs to be clearly distinguishable from the thesis itself, standing by its own merits. This work should neither be part of the thesis, but serve as an starting point that can be referred within the thesis bibliography.	
Study Goals	After the course the student should be able to: - formulate a clear problem definition with research questions; - independently carry out an assignment; - document the results of the assignment.	
Education Method	There are no lectures scheduled for this course. Instead, an individual assignment is carried out. The assignment should be supervised by a staff of the MSc Geomatics programme at TU Delft, and has to be approved before the start by the course coordinator of GEO5010. The coordinator should receive a copy of the proposed assignment before the actual start of the work, and of the final deliverable at the end of the assignment. To start the project, a half-page summary containing the following is required: - name student - title of project - main objectives and deliverables - planned timeframe	
Assessment	The marking of an assessment is made by the supervisor of the project and the coordinator of the course (in order to guarantee uniformity of marking in relation to the level of difficulty of different assignments). There is in theory no time limit for an assignment, that is a student can start for instance during the summer and work part-time on the project for a few months. An assignment can be handed-in at any time during the year.	
Special Information	The maximum marking period is 10 work days.	
Remarks	Brightspace is not used, all the information is there: https://3d.bk.tudelft.nl/courses/geo5010/	
Period of Education	no restriction, a project can start/finish at any time of the year	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	Mr.dr. H.D. Ploeger	
Education Period	3	
Start Education	3	
Exam Period	3	
	4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After this course the student is able to: - Explain the basis of land administration (why is this needed?, the different components: survey, spatial data, legal framework documents, etc.); - Describe and explain the underlying principles and technology of Land Administration Systems (LAS); - Describe and assess the role of actors within a specific system; - Explain the significance of LAS for sustainable development and take a personal and critical position on this; - Assess, based on a systems approach, the practical value of a specific (national) LAS; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Motivate the importance of standards in general and specifically the ISO standard 19152 (Land Administration Domain Model); - Make proposals for improvement of a specific LAS in general terms. - Explain the increasing need for future developments such as 3D Cadastres.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	Written exam (3 hours).	
Period of Education	Quarter 3.	

GEO5014		Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro		
Instructor	Mr. C.A. León Sánchez		
Responsible for assignments	Dr. G. Aguiaro		
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)		
Education Period	1		
Start Education	1		
Exam Period	1		
Course Language	English		
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent		
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.		
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.		
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.		
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)		
Assessment	- Graded and nongraded assignments during the quarter - Group-wise final project consisting of a written report and an oral presentation at the end of the course.		
Period of Education	Quarter 1 of the second year of MSc Geomatics		
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. (Generally), the course takes place in the afternoon on Tuesdays, Wednesdays and Thursdays.		
Minimum number of participants	15		
Maximum number of participants	30		

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	1) Understand the fundamental requirements for urban energy modelling and for micro-climate simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Depending on the specific application, implement the required computation procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool; 4) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem; 5) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists on hands-on assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

GEO5016	Geomatics in practice	10
Course Coordinator	F.M. Welle Donker	
Responsible for assignments	F.M. Welle Donker	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	You should have completed at least 30 ECTS of core Geomatics Programme MSc 1 and MSc 2. This is a non-negotiable condition.	
Course Contents	To apply Geomatics knowledge and skills obtained in the first year of Geomatics for the Built Environment in practice at academic level.	
Course Contents Continuation	There is no formal start date or end date, you can start any time provided you have completed at least 30 ECTS of Geomatics core subjects. The internship should last at least 280 hours, i.e., seven weeks on full-time basis, or an equivalent period on part-time basis, e.g., 14 weeks @ 20 hours/week. BEFORE YOU START: When you have found an internship at a suitable academic level, you should submit an Internship Proposal form (maximum 800 words) via Brightspace, stating: 1. Your name, study number, email address and completed geomatics core courses 2. Short description of the internship organisation 3. Name, affiliation and qualifications of the supervisor at the internship organisation (internship coordinator) 4. Name, affiliation of proposed Geomatics Supervisor 5. Topic of the work you will carry out, objected achievements and deliverables to the internship organisation 6. Motivation why you want to carry out this work and indication how it will contribute to your geomatics knowledge and skills 7. Indication of the academic level of the work you will carry out 8. Time planning Please involve your Geomatics Supervisor when you draft your internship proposal. The Geomatics Supervisor may be any person connected to the Geomatics programme with sufficient knowledge of your chosen topic. When you submit your Internship Proposal via Brightspace, please inform the GEO5016 Coordinator by e-mail of your submission. The GEO5016 Coordinator will assess your Internship Proposal for suitability for the Geomatics programme. A GO will be granted when the internship Coordinator is satisfied that the proposed topic and activities are suitable for the Geomatics Programme and at MSc level, i.e., there has to be an academic challenge. This means that you have to work towards tackling a problem, or work with complex and heterogenous data, or develop a model or something similar, either individually or as part of a team. So, just carrying out activities according to fixed protocols without your own contribution is not sufficient. Without a GO, you are not allowed to start this internship. Do not forget to enrol in GEO5016 Geomatics in Practice, Aanvangsblok: JAAR. WHEN YOU RECEIVE A GO: Submit a TUDelft Internship Agreement, signed by you and your Workplace Supervisor, to J.vanOs@tudelft.nl with a cc to the GEO5016 Coordinator. The Internship Agreement form can be downloaded from https://www.tudelft.nl/en/student/faculties/a-be-student-portal/education/forms/. DURING YOUR INTERNSHIP: Carry out your internship activities and absorb as much practical experience as you can! AFTER YOUR INTERNSHIP: Submit your Internship Report to your Geomatics Supervisor with a cc to the GEO5016 Coordinator no later than 4 weeks after completion of your internship. Make an appointment with your Geomatics Supervisor to discuss your report and to gain an assessment mark.	
Study Goals	To gain experience in a company or governmental organisation and learning to explore knowledge and skills in a practical environment. After this course you will be able to apply your newly acquired knowledge at skills to your future work.	
Education Method	Individual work: Internship Report (max 5,000 words) according to template provided via Brightspace.	
Practical Guide	You can start any time you want, there is no formal start date or end date, provided you have completed at least 30 ECTS of Geomatics core subjects. This is a non-negotiable condition. The internship should last at least 280 hours, i.e., seven weeks on full-time basis, or an equivalent period on part-time basis, e.g., 14 weeks @ 20 hours/week. BEFORE YOU START: Find an internship at a suitable academic level. Find a Geomatics Supervisor. Submit an Internship Proposal form to the GEO5016 Coordinator to receive a GO. Without a GO, you are not allowed to start this internship. Enrol in GEO5016 Geomatics in Practice, Aanvangsblok: JAAR. WHEN YOU RECEIVE A GO: Submit a TUDelft Internship Agreement (available via https://www.tudelft.nl/en/student/faculties/a-be-student-portal/education/forms/) signed by you and your Workplace Supervisor, to J.vanOs@tudelft.nl with a cc to the GEO5016 Coordinator. DURING YOUR INTERNSHIP: Carry out your internship activities and absorb as much practical experience as you can! AFTER YOUR INTERNSHIP: Submit your Internship Report to your Geomatics Supervisor with a cc to the GEO5016 Coordinator no later than 4 weeks after completion of your internship. Make an appointment with your Geomatics Supervisor to discuss your report and to gain an assessment mark.	
Prerequisites	You should have completed at least 30 ECTS of core Geomatics Programme MSc 1 and MSc 2. This is a non-negotiable condition	
Assessment	The final product of your internship is a report. Based on the report, your written reflection on your results, oral elucidation and feedback from your internship supervisor you will be graded by your Geomatics supervisor (fail; 6; 7; 8; 9;10). The criteria on	

	which you will be assessed are: initiative and creativity; insight in the goals and work approaches of the company/organization; quality of the deliverables; communication with your internship supervisor. The assessment is based on the Rubric for assessment of GEO5016, available via Brightspace
Enrolment / Application	Please enrol in GEO5016 Aanvangsblok: JAAR
Period of Education	There is no formal start date or end date. The internship should last at least 280 hours, i.e. seven weeks on full-time basis , or an equivalent period on part-time basis, e.g. 14 weeks @ 20 hours/week.

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After finishing this course, the students will have gained the theory of commonly used machine learning techniques and the skills to apply them for processing geospatial data. Specifically, the students will be able to: - understand and explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, dimensionality reduction, overfitting, training, validation, cross-validation, learning curve, and regularization); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, random forest, and neural networks); - collect and preprocess data (e.g., labeling, normalization, feature selection, augmentation, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification or semantic segmentation); - analyze and evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what' and a brief reflection on how the feedback received from others improves the work; - Strict deadline: 10% deduction per day late, no more accepted after 3 days.	
Prerequisites	- Linear algebra - Calculus - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of three group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e.,	

	- Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 0.5). - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Both assignments and the final exam have to meet the minimum requirement (i.e., 5.5) to pass the course. A total of 6.0 or above is necessary to pass the course.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Concept Schedule	Tuesday and Thursday

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Nearly eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day,. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in the data-driven city of 2050 to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in the city of tomorrow: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, students are able: - To understand the relevance of applied ethics in relation to the design and governance of the data driven city - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with a short oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes), a poster or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. During the course the student must answer one or more short (closed and/or open) questions after reading each of the papers included in the syllabus. The answers given will be assessed with a pass or fail. Answering all questions with a pass is a necessary condition to complete the course.	
Period of Education	Quarter 4	
Maximum number of participants	20	

Year	2022/2023
Organization	Architecture and the Built Environment
Education	Master Geomatics

Electives, courses listed below are recommended to be chosen as electives

AP3421	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. L. di Carlo	
Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics.	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic algorithms and quantum algorithmic techniques; - basic quantum error correction; - simple physical implementations of qubits.	
Study Goals	Motivation: Quantum information is the future of computing and communication. Quantum computers offer exponential speedup over any classical computer. Similarly, quantum communication offers many advantages, including the ability to create secure encryption keys where security rests only on the laws of nature. Synopsis: This class will teach you the fundamental principles of quantum information. You will learn essential concepts that distinguish quantum from classical devices. You will learn about quantum bits and the quantum operations and measurements that can be performed on them. You will learn the basic techniques used in quantum algorithms, and examine basic examples of such algorithms. You will also take the first step in understanding how a quantum bit can be physically implemented. Aim: To learn the fundamental concepts underlying quantum computation and communication systems.	
Education Method	3 hours of lecture, 1 hour tutorial per week.	
Literature and Study Materials	The main reference textbook for the course is Nielsen and Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% in class quiz, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	
Permitted Materials during Tests		
Continuing Courses	This course forms part of the curriculum on Quantum Technologies offered at TU Delft, which at present consists of: AP3421 Fundamentals of quantum information AP3421-PR Quantum Information Project CS4090 Quantum communication and cryptography AP3432 Quantum Hardware 1 - Theoretical Concepts AP3442 Quantum Hardware 2 - Experimental State of the Art EE4575 Electronics for quantum computation	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	

CESE4055	Ad-hoc Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ----- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. ----- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) - o Definition of ad-hoc networks - o Comparison with infrastructure based systems - o Typical applications - o Advantages and challenges - o Radio technologies for ad-hoc networks - o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks - o Propagation models - o Topology models based on graph theory - o Degree and hopcount - o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) - o Introduction to MAC protocols - o Issues and design goals - o Classification - o Directional, multi-channel MAC protocols - o Energy efficiency in MAC protocols - o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) - o Flooding - o Node discovery, neighbour discovery - o Route establishment - o Topology maintenance, localisation - o Proactive, reactive and hybrid routing - o Typical protocols - o Energy efficiency in routing - o Broadcast and multicast - o Effects of mobility on connectivity and capacity - o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks - o Wireless sensor networks (Chapter 12 of the textbook and papers) - o Cooperation (Reference papers) - o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) - o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

textbooks, research papers, etc).
3. Some recent journal papers

4. Optional Reference Books

- 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms)
- 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library.
- 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library.
- 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer.
5. OPNET/ns-2 web pages, tutorials and video lectures

Books

Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.

Assessment

1. There will be totally three written/oral tests/examinations for this course (including the final exam).
2. The students will carry out a project in a group and submit a short report.
3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis.

Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class.

Breakup:

2 Tests + Final Exam = 55%

Project 40%

Self assessment + Reflection 3%

Activities on Feedback Fruits or any Online platform 2%

Resit: written/oral exam [+ carry over from projects].

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Changes due to COVID19 in 2021 is below;
in case there is a similar situation, we use the following methods.

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There are three chapters

Chapter 1: Network modelling

Chapter 2: MAC protocol

Chapter 3: Routing

+

Lab: Simple experiments using laptops (individual) + simulations (group-wise)

Marks Breakup

Homework/Assignments/Group works

Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks

Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks

Part B1: Individual experiment + report

Part B2: Group-wise simulations + report + demo

Part B1 + Part B2 - 50 Marks

Part C: Self-assessment + peer activities -- 5 marks

Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit.

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(More information will be given in the first class)

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Permitted Materials during Tests

Different conditions for different test/exams.

Conditions will be informed 1 week before the exams/test.

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands on course for self motivated students. Students work in a group of two (max 3) usually. Students are expected to have sufficient programming and hardware development skills. The course is project oriented thus students are expected to deliver a working model with demonstrations. Preference will be given to Q5 students. Q1 students are discouraged unless otherwise explicitly allowed. ----- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. ----- This course is done in a group of two (max 3). Instructor/TA will help connecting a potential groupmate in the first class using Brightspace/Feedback Fruits. ----- Activities: How to in Groups? ----- Part 1: Seminar We guide the groups to select a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID19 related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their own responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/raspberry pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki based networks or Open testbeds projects. In these cases, online group activities are possible. NOTE: In some extreme cases, Instructor/TA may allow one person project after assessing the situation and after discussing with the student. ----- Supply of Hardware: ----- 1. Instructor/TA is able to provide take home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	
Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc.	

	Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Circuits Group work Programming Project Software
Judgement	30 Marks for Seminar 60 Marks for project, demonstration, Q/A 10 Marks for making a paper that could be submitted as a manuscript for review to a conference/journal. Bonus marks are offered for excellence in various components during the course. That would add to the total marks not exceeding a max of 100.
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CESE4075	Supercomputing for Big Data	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming. Experience in Java, Scala or Python is a plus due to their importance in the big data ecosystem.	
Course Contents	Big data is one of the hottest technology fields today, used to describe large and complex data sets that are difficult to process using traditional data processing systems. In this course, we will introduce the students to the most important concepts of big data analytics and the available tools and systems used to manage and process it. In a series of labs, the students will be working with a number of different big data applications, addressing such aspects as implementing big data algorithms using Spark and Hadoop on high performance computational systems, as well as using higher-level languages and system management tools to guide and monitor the performance of these systems. The students will have access to a big data cluster to evaluate the effectiveness of their implemented solution in practice.	
Study Goals	By the end of this course, the students will be able to - use basic big data processing systems like Hadoop - implement parallel algorithms using the in-memory Spark framework, and streaming using Kafka - use libraries to simplify implementing more complex algorithms - identify the relevant characteristics of a given computational platform to solve big data problems - utilize knowledge of hardware and software tools to produce an efficient implementation of the application	
Education Method	Lab assignments and reading material provided via Brightspace and online on the CognitiveClass.ai site	
Assessment	The students will be evaluated based on the quality of their implementation and the report they submit describing their solution approach. The course has three lab assignments: L1, L2 and L3. The final grade of the course will be evaluated as $\text{Grade} = 0.5 * L1 + 0.3 * L2 + 0.2 * L3$ Disclaimer: information may change depending on the developments around coronavirus.	

CIE4510-20	Climate Change: Science and Ethics	4
Responsible Instructor	Dr. M. Vizcaino	
Instructor	Dr. R.E.M. Riva	
Instructor	Prof.dr.ir. B. Taebi	
Instructor	Dr. S.L.M. Lhermitte	
Instructor	F. Glassmeier	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Climate change is one of the most profound and complex issues affecting our society and economy today. Many scientists argue that there are too many variable factors to effectively see the big picture, while other scientists who believe human activity is to blame for global warming are ready to outline specific actions to prevent more damage. Skeptics believe that climate change is part of the natural global progression and that human activity will neither worsen nor improve our situation. Those who are in favor of a global effort to reverse climate change believe that current climate models are an useful tool to estimate future warming and argue that the uncertainty surrounding this threat is no excuse for inaction. Skeptics in turn argue that scientists who want to attract attention to themselves, who want to attract great funding to themselves, have found a way to scare the public by making things bigger and more dangerous than they really are. Despite continuing uncertainties about the detailed linkages, extreme weather events are increasingly being attributed to human interference, and greater emphasis is emerging on the need to prevent and to adapt to climatic changes. The course provides an introduction to the basic physics of the climate system, how climate has changed in the past and how climate will change in the future. The focus is on the energy balance of the climate system and how this balance is affected by greenhouse gases and aerosols; the physical processes in the atmosphere and oceans that shape the climate; the response of the oceans, ice sheets and glaciers to global warming; the evidence for past and present climate change; climate models and model uncertainties; climate predictions. A second focal point of the course is the broader societal and ethical aspects of climate change. In particular, we will focus on past emissions and responsibilities, implications of global warming on human safety and security, the distribution of burdens and benefits, emission rights, international justice and intergenerational justice. Syllabus: - Introduction to climate physics - Records of past Earth's climate - Radiative heat transfer - Atmospheric circulation - Clouds, aerosols, and climate - The carbon cycle - Forcings and feedbacks in the climate system - Climate change and sea level rise - Ice in the Climate System - Climate modeling and predictions	
Study Goals	After completing of the course, the student is able to - explain, make calculations and analyse model results on the basic physics of the climate system regarding energy budget, climatic effects of aerosols and clouds, atmospheric circulation, and the distinction between forcings and feedbacks. - explain, exemplify and make basic calculations on how the climate responds to anthropogenic carbon and aerosol emissions, and land use change. - describe how future climate change is predicted and explain the sources of model uncertainties - calculate future atmospheric greenhouse gas concentrations for different emission scenarios - describe, explain and quantify the impacts of climate change (e.g. sea-level rise) as well as potential mitigation strategies - explain and apply techniques to detect and attribute climate change from past climate and observational records - assess, propose and defend engineering solutions to the climate change problem taking into account their social and ethical implications	
Education Method	Video lectures (edx course 12.340x Global Warming Science) In-class question-and-answer and feedback sessions related to the video lectures In-class lectures to address specific topics not covered by the video lectures Climate lab	
Assessment	essay + exam	
Tags	Ethics	
Expected prior Knowledge	BSc diploma	
Academic Skills	Critical thinking Written and oral communication Ability to speak and listen Teamwork Conceptual thinking Interpretation Moral awareness & sensitivity Judgemental skills Debating and discussion	
Literature & Study Materials	1. video lectures 2. in-class lectures 3. Kump, Kasting & Cane, The Earth System 4. Farmer & Cook, Climate Change: A modern synthesis 5. IPCC Fifth Assessment Report (AR5), http://www.ipcc.ch/report/ar5/ 6. Online material for further reading, e.g., - US EPA homepage, http://www.epa.gov/climatechange/index.html - Understanding climate change, http://www2.ucar.edu/news/backgrounders/understanding-climate-change-global-warming - NASA global climate change, https://climate.nasa.gov - Online reports of the US National Academic Press (NAP)	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE):	

	https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Pocket calculator
Collegerama	No

CIE4522-15	GPS for Civil Engineering and Geosciences	4
Responsible Instructor	Dr.ir. H. van der Marel	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. A.A. Verhagen	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is offered only as a resit opportunity for Geoscience and Remote Sensing students who were unable to take the course last year. Global Navigation Satellite Systems (GNSS), such as GPS, have revolutionized positioning and navigation, and resulted in novel applications. This course addresses the principles of GNSS, and covers the main applications in Civil Engineering and Geosciences: 1. Fundamental principles of Global Navigation Satellite Systems (GNSS) This part covers the basic principles and components of GNSS; the American GPS, Russian Glonass, European Galileo, and Chinese BeiDou system. You will learn how a basic GNSS receiver – a handheld, mobile phone or car navigation receiver – computes the position and get a basic understanding of the error sources, accuracy and limitations of GNSS. Topics are: space segment, user and control segment, overview of systems, signal structure, clocks, receiver architecture and signal processing, pseudorange measurement, broadcast and precise ephemeris, signal propagation error sources, multipath, standalone or single point positioning, estimation of position, velocity and time, smartphone and car-navigation applications, augmentations systems (SBAS). 2. High-precision positioning and navigation This part focusses on methods to improve the accuracy of standard GPS positioning down to the millimeter level. You will discover the techniques that make millimetre GNSS possible: interferometric measurement principle, differential/relative positioning with two (or more) receivers, carrier phase measurement, mathematical models with Single- and Double-Differences, carrier-phase ambiguity resolution, parameter estimation for dynamic processes (e.g. a moving receiver), Kalman-filter. 3. GNSS infrastructure, high-precision implementation aspects, and applications In this part you will learn how to apply the high-precision techniques in an operational setting for civil engineering, land-surveying, mapping, hydrography, transport and traffic applications. Different operational scenarios will be covered: fully autonomous, setting up your own base stations, or make use of service providers, network RTK services, and how to make use of permanent GPS networks and the International GNSS Service (IGS) for Precise Point Positioning. Real-time operation and tele-communication aspects are also covered, as well as post-processing applications and available free services for Precise Point Positioning. 4. GNSS for Geodynamics and Remote Sensing In this part GNSS application for geodynamics studies and remote sensing are covered. You will learn how GNSS is used to measure plate tectonics, deformations at the plate boundaries and volcanos, land subsidence and global isostatic adjustment and the role of global networks of GNSS receivers (IGS) for maintaining the international reference frame and determination of Earth Rotation parameters. In the part on GNSS remote sensing you will learn how GNSS can be used to sense water vapour estimation for weather forecasting and climate research, estimate Total Electron Content and electron density of the ionosphere for space weather applications, and do radio occultation with GNSS receivers on board Low Earth Orbiting satellites for atmospheric profiling, GNSS-reflectometry for ocean altimetry, ocean state and soil moisture measurements. 5. GNSS processing workshop The last two weeks of the course students will be involved in some hands-on GNSS data processing. The teachers will select together with the students one or more suitable datasets and softwares; 3-4 students will work on the same dataset and present their results in a presentation to the plenary.	
Study Goals	After successfully completing this course, a student is able to: - explain the basic ranging, timing and positioning principles of GPS, - apply standalone positioning and obtain a position solution given a set of pseudorange measurements, - evaluate the effects of error sources on GPS ranging and positioning and propose effective methods to mitigate or eliminate the resulting errors, - set-up the observation model for different relative positioning scenarios, - show how GPS can be used for a given application in application in civil engineering or geosciences, - do simple GNSS data processing and analysis.	
Education Method	Two contact hours per week for question and answer. The students are required to watch the recorded lectures of last year before the weekly sessions.	
Assessment	Students will be graded primarily through a series of assignments (toets01). The assignments can be done individually, or in groups of two students. Next, there is a short (1.5 hour) written open-book exam at the end of module 4 (toets02), and an oral presentation on results of the GNSS processing workshop (toets03). The weighting is 50% assignments (toets01), 25% for a short (1 hour) written open-book exam at the end (toets02), and 25% for the GNSS workshop results and oral presentation (toets03).	
Elective	Yes	
Tags	Algorithmics Building & Spatial Development Diverse Embedded systems Fieldwork Geo Engineering Information & Communication Physics Rail & Road Engineering / Planning Signals and Systems Small groups Technology Transport & Logistics Water Engineering Water management	
Expected prior Knowledge	This course provides an introduction to GPS. Prior knowledge on related subjects such as parameter estimation theory, as	

Academic Skills	covered in CIE4611 Geo-measurement processing, and basic measurement principles and reference systems, as covered in CIE4606 Geodesy and Remote Sensing is recommended. Relevant material can be provided on request.
Literature & Study Materials	literature study, critical thinking, research abilities, problem solving, discussion with peers, presenting an application idea, application to practical problems
Judgement	In addition to the lecture-slides, a set of hand-outs on essential topics of this course, supplemented with selected papers for further reading and background.
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Collegerama	During the exam the students are allowed to consult the material published on Brightspace and use their own written notes.
	No

CIE4601	Physics of the Earth and Atmosphere	5
Responsible Instructor	Dr. S.R. de Roode	
Instructor	Prof.dr. L.L.A. Vermeersen	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic mathematical skills. Experience with Matlab is desirable.	
Course Contents	This course will present a common introduction to the major Earth system components, including the atmosphere, ocean and solid earth. Basic physical principles will be discussed and numerous examples from observations will be shown to provide the students with an elementary background in some key phenomena that occur on Earth. An example of some questions that will be addressed in the lectures are: * why do we need the greenhouse effect for a livable planet? * why does warm air rise and why can't we fly in gliders during nighttime? * how does the pressure in the atmosphere or ocean vary with height? * why does the wind approximately flow along lines of equal air pressure? * why is the weather typically dry and sunny in an high pressure system? * how can we use satellite altimetry to estimate the flow velocity in the ocean? * how does one model transport of heat and mass in an earth-system model? * what drives plate motion? * which locations on Earth are most vulnerable to earthquakes and volcanoes? * how much is the Earth's shape changing on a daily basis? * what happens below the Earth surface? * what is the influence of the solid Earth on sea level rise? * what is the relation between rotational variations of the Earth and geophysical processes? An introduction to the Earth System ES1. Components (spheres: atmosphere, hydrosphere, lithosphere, biosphere, cryosphere) and interactions (internal/external) ES2. Energy balance of the Earth System An introduction to the atmosphere A1. Atmospheric composition and vertical structure A2. Radiative transfer and the earth-atmosphere energy balance, greenhouse effect A3. Key physical laws: Conservation equations: mass, heat, moisture and momentum Thermodynamics: gas law, Clausius-Clapeyron A4. Vertical stability of the atmosphere (troposphere) A5. Water in the atmosphere Large-scale dynamics of the atmosphere and oceans AO1. Equations of motion with coriolis force (Hadley cells...) AO2. Ocean vertical structure (and bottom pressure) AO3. Geostrophic currents, thermal wind relationship , inertial oscillations AO4. Currents with friction; Ekman layer; Ekman transport An introduction to the Solid Earth SE1. Earth structure, lithosphere structure, heat flow SE2. Plate tectonics, mantle convection, hot spots SE3. Gravity and isostasy SE4. Tidal potential and solid earth tides SE5. Surface loading and elastic deformation SE6. Viscoelastic deformation	
Study Goals	To get acquainted with the most important physical processes in the Earth system. AO1. Being able to apply the energy balance to a planet with a single-slab atmosphere. AO2. Being able to apply the hydrostatic balance equation, the gas law and the Clausius-Clapeyron equation. AO3. Being able to calculate the vertical stability of the atmosphere. AO4. Being able to express the conservation equations for momentum, heat and mass in a dimensionless form and to be able to assess the dominant terms that, for example, lead to the geostrophic balance equations. Understand the meaning of the Reynolds and Rossby numbers. AO5. Being able to derive and apply the relation of Ekman pumping. AO6. Being able to derive and apply the relation between the wind field and the large-scale ocean gyres. SE1: Being able to understand solid-earth geophysical processes in general and plate tectonics in particular, and their relations with the internal structure of the Earth. SE2: Being able to understand the relation between solid earth deformation, gravity and sea level change. SE3: Being able to determine rotational variations of a deformable Earth and their relationships with geophysical processes and tidal variations.	
Education Method	15x2 lecture hours. Six sets of exercises will be provided which will take about ~ 10 hours per set. Experience with Matlab (or IDL, Python etc.) is desirable. Otherwise it is expected that the student will take an effort to learn its basic features. The students will have the opportunity to work on the exercises under the supervision of the instructors (6x2 hours). The exercises will be graded and will determine 50% of the final grade.	
Computer Use	Matlab	
Books	"Atmosphere, Ocean and Climate Dynamics" by Marshall and Plumb, This Dynamic Earth by Kious and Tilling (open source), handouts, lecture notes.	
Assessment	ass. + exam	
Expected prior Knowledge	linear algebra, calculus, classical mechanics, programming (either Matlab, Python or a comparable package)	

Academic Skills	coding
Literature & Study Materials	The book 'Atmosphere, Ocean and Climate Dynamics' by Marshall and Plumb covers basically all the Atmosphere and Ocean topics and may be purchased.
Judgement	The final grade is based on compulsory exercises and a written exam (50-50). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	scientific calculator, relevant equations will be provided in a separate formula sheet
Collegerama	No

CIE4602	Cryosphere: Remote Sensing and Modelling	4
Responsible Instructor	Dr. M. Vizcaino	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	1.The Cryosphere and Climate Change a.Overview of Cryosphere components: glaciers, ice sheets, sea-ice, seasonal snow cover, frozen ground b.Global and Polar climate change c.Sea-ice change: processes, observations and models 2.Snow and ice surface processes (accumulation, melt, refreezing, sublimation) a.From snow to ice; classification of snow and ice covered surfaces b.Polar meteorology: processes and observations c.The Surface Energy and Mass budget: radiation fluxes, melt, albedo d.Optical and near-infrared remote sensing techniques applied to snow and ice surfaces e.Modeling glacier and ice sheet surface processes 3.Glacier flow a.The flow and temperature of ice masses b.Remote sensing of ice flow c.Models of ice flow 4.Ice sheet and Climate coupling a.Processes and observations of mass loss of the Greenland and Antarctic ice sheets b.The glacial cycles of the last three million years c.Monitoring ice sheet calving fronts, elevation change, and thickness d.Predicting future Greenland and Antarctic ice sheet evolution	
Study Goals	This is a course on the physics of the parts of the Earth System that contain frozen water (the seasonal snow cover, sea-ice, glaciers, and the Greenland and Antarctic ice sheets), on their interaction with the physical climate system, and on the techniques to observe these parts of the Earth remotely. Special emphasis will be given to glaciers and the Greenland and Antarctic ice sheets, that are currently losing mass at an accelerated pace and becoming a major contributor to sea level rise. After this course, students are able to: identify the mechanisms of current mass loss from glaciers, shrinkage of the Arctic sea cover, and reduction of the seasonal snow cover connect ice and snow changes with global climate change identify the impacts of snow and ice change on human societies and the environment apply optical, infrared and other remote sensing techniques to monitor ice and snow change combine observational and modeling tools to monitor and predict ice and snow change identify current challenges in the observation and modeling of ice and snow surfaces	
Education Method	Materials: lectures, videos, numerical exercises and problems, guided assignments, practical classes. The student will use several remote-sensing techniques to monitor glacier, snow-cover, ice sheet and/or sea-ice change. He/she will use models to simulate and/or analyze glacier flow, surface melt, global and local climate change, and/or coupled ice-sheet/climate evolution.	
Assessment	The Assessment will be done from Assignments (50% of the final mark) and a written exam (50% of the final mark). Assignments may be done in small groups. The Exam will be a mixture of short theoretical questions, multiple choice and numerical exercises. Minimum grade to pass the course is 6.0 for the total, and 5.0 for each of the components (Assignments and Written Exam)	
Elective	Yes	
Tags	Physics Research Methods Technology	
Expected prior Knowledge	Physics at the Bachelor Level.	
Academic Skills	Recommended: CIE4613 Climate Change and ethics and CIE4601 Physics of the Earth and atmosphere.	
Literature & Study Materials	Programming, data manipulation, data visualization (e.g. QGIS, ArcGIS, MATLAB) Required: - Slides from lectures in class as posted in BrightSpace - Numerical exercises and problems posted in BrightSpace Recommended: - The Physics of Glaciers, Paterson & Cuffey - "Remote Sensing of snow and Ice", Rees, 2006, CRC press - "Remote sensing of the Cryosphere" by Marco Tedesco - Polar remote sensing, Masson & Lubin http://link.springer.com/book/10.1007%2F3-540-30785-0 http://link.springer.com/book/10.1007%2F3-540-30565-3 - Fifth Assessment Report of the International Panel of Climate Change, chapters on Observations of the Cryosphere and on Sea Level.	
Judgement	Final mark: 50% from Assignments 50% from written exam The passing mark is 6.0. Both parts need to be passed with a minimum of 5.0	

	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Pocket calculator
Collegerama	No

CIE4603-16	Geo-signal Analysis	6
Responsible Instructor	Dr. M.A. Schleiss	
Instructor	Prof.dr.ing. R. Klees	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Basics in programming (e.g., Matlab, Python or R) - Basics in statistics and probability theory - Basics in Euclidean and spherical geometry - Linear algebra (vectors, matrices, solving linear systems of equations) - Calculus (limits, derivatives and integrals of real-valued functions) - Complex numbers, trigonometry	
Course Contents	The course consists of two parts: Geospatial analysis (Dr. Marc Schleiss) and Spectral Methods (Prof. Roland Klees), running in parallel to each other. Contents of "Geospatial analysis": - Deterministic interpolation: weighted distance interpolation, nearest neighbors, triangular and bilinear interpolation - Structural analysis: random fields, second-order-stationarity, covariogram, variogram, nugget, sill, range, anisotropy, variogram estimation and modeling - Stochastic interpolation: best linear unbiased estimator, kriging equations, ordinary kriging, kriging with local neighborhoods, screening and smoothing effects. - Merging: weighted distance interpolation merging, cokriging, kriging with external drift - Classification: binary decision trees, k-nearest neighbors, variance-bias tradeoff, k-fold cross-validation, feature selection/reduction, principal component analysis. Contents of "Spectral Methods": - Fourier transforms of functions on the continuous domain: Fourier series, Fourier integral, two-dimensional transforms in Cartesian space. - Convolutions and windows on the continuous domain: convolutions of non-periodic functions; convolutions of periodic functions, filters on the real line and plane, window functions. - Transforms, convolutions, and windows on the discrete domain: infinite sequences (Fourier transform, discrete convolutions, aliasing); periodic sequences (discrete Fourier transform, discrete cyclic convolution, aliasing); discrete filters and window. - Correlation and power spectrum: correlation of finite energy functions, correlation of finite-power functions, correlation of periodic functions, stochastic processes, estimation of the covariance and power spectral density.	
Study Goals	This course introduces the basic concepts, tools and techniques for analyzing data in geosciences and remote sensing: deterministic interpolation, structural analysis, kriging, classification, Fourier transforms, spectral analysis, filtering, and smoothing. The goal is to gain a good theoretical and conceptual understanding of these techniques by applying them to real-world problems and datasets derived from ground-based, airborne and space-borne sensors. After this course the student should be able to assess the structure and information content of spatially and temporally distributed data. Notably, the student will be able to: (1) recognize data of different types and dimensions (2) detect patterns and quantify spatial/temporal structures in noisy signals (3) apply geostatistical and spectral methods to the processing of real data using a programming environment like R, MATLAB or Python (4) evaluate the accuracy and computational complexity of competing models and justify a choice for a certain analysis, interpolation, filtering or classification method (5) Present findings in a precise and organized way and critically assess the quality of the obtained results.	
Education Method	This course consists of a mix of lectures, supervised instruction (practicals) and self study: The average work load (on a weekly basis) is given by: - 4 hours of lecturing - 4 hours of supervised instruction (practicals) - 4 hours self study, doing exercises and solving problems. - 6 hours working on graded assignments and reports. --- UPDATE --- Due to the transition to the new MSc program, this course will be offered for the last time in 2022-2023. The lectures will be replaced by Q&A sessions and the recordings from previous years will be uploaded for self-study. Students will be given the opportunity to redo the assignments and the exam. Please contact the responsible teacher for more information.	
Assessment	ass. + exam	
Contact	Dr. Marc Schleiss, m.a.schleiss@tudelft.nl CITG building, room 2.18	
Expected prior Knowledge	Linear algebra, calculus, basic statistics, basic programming	
Academic Skills	Presenting (Assignment results will be presented in classroom) Academic writing (assignments) Collaboration (Assignments will be done in groups of two)	
Literature & Study Materials	Lecture notes and slides will be provided. Optional background reading and references: - "A Practical Guide to Geostatistical Mapping", Tomislav Hengl, 2009 (Open Access) - Multivariate Geostatistics : an Introduction with Applications, Hans Wackernagel, 2003 - An Introduction to Statistical Learning, Gareth James, Springer 2017 (Open Access) - Spectral methods in Geodesy and Geophysics, C. Jekeli - Mathematical Principles of Remote Sensing, A.S. Milman - Digital Processing of Random Signals, B. Porat	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

Permitted Materials during Exam	A simple pocket calculator without memory is allowed.
Collegerama	No

CIE4606	Geodesy and Remote Sensing	5
Responsible Instructor	Dr.ir. D.C. Slobbe	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	After defining the discipline of Geodesy and its role in Earth sciences, we explore the reference systems and reference frames used in Geodesy (both for time and positions). In particular, we will study the way they are defined and how their realizations are obtained in practice. Second, we will classify the two main positioning methods by the type of measurements used to compute a position solution, and work out in detail positioning using Global Navigation Satellite Systems (GNSS). Next, we treat the determination of the Earth's gravity field and figure. Starting with the fundamentals of potential theory, we will derive Geodetic Boundary Value Problems and outline the strategy to solve them using gravity measurements performed at the Earth's surface. In a separate lecture, we will discuss the meaning of the word "height" in view of what we learned about the figure of the Earth. The second part of this course treats the physical principles of remote sensing based on electromagnetic radiation. First, we will explore the different forms of interaction of electromagnetic radiation with the Earth's surface and the Earth's atmosphere. After that, we will compare the remote sensing measurement techniques (i.e., electro-optical systems, ranging systems, and radar) in terms of underlying physical principles and limitations. Finally, given that we want to study a particular phenomenon or process on Earth, we will learn how to select an appropriate survey platform and, if applicable, orbit design by taking into account requirements regarding target accuracy, spatial coverage, and spatial/temporal resolution.	
Study Goals	At the end of this course you will be able to: 1. Explain the role of Geodesy in Earth Sciences; 2. Analyze the reference systems used in Geodesy in terms of: (i) definition and (ii) the ways their realizations are obtained; 3. Apply coordinate transformations among the reference systems used in Geodesy; 4. Explain how positions can be obtained from GNSS code and carrier phase measurements; 5. Summarize the procedure how to derive the figure of the Earth from gravity observations; 6. Describe the different forms of interaction of electromagnetic radiation with the Earth's surface; 7. Describe the different forms of interaction of electromagnetic radiation with the Earth's atmosphere; 8. Compare the remote sensing measurement techniques in terms of underlying physical principles and limitations; 9. Select an appropriate survey platform and, if applicable, orbit design to acquire data from a particular phenomenon or process on Earth by taking into account requirements regarding target accuracy, spatial coverage, and spatial/temporal resolution.	
Education Method	Lectures, in-class assignments, in-class demonstrations, and take-home assignments	
Assessment	ass. + exam	
Expected prior Knowledge	The expected prior knowledge is the same knowledge required for enrolling in the MSc-track Geoscience and Remote Sensing, which can be found at http://www.tudelft.nl/en/study/master-of-science/master-programmes/applied-earth-sciences/msc-programme/tracks/geoscience-and-remote-sensing/programme/prior-knowledge/	
Academic Skills	Thinking (critical, analytical); Interpretation; Problem solving	
Literature & Study Materials	1) Lecture slides; 2) Some chapters of selected textbooks (provided via BrightSpace or available online); 3) Rees, W. G. (2013). Physical principles of remote sensing. Cambridge University Press.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Formula sheet (will be provided) and pocket calculator.	
Collegerama	No	

CIE4607	Ocean topography and sea-level change	4
Responsible Instructor	Dr. R.E.M. Riva	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The ocean surface is continuously changing due to processes acting at different spatial and temporal scales. Sea level is affected by sub-daily tides, by multi-yearly changes in ocean circulation, but also by climate change and by long-term geophysical processes. For instance, global mean sea level is currently rising at a rate of about 3 mm/yr and is expected to accelerate during the coming decades, potentially rising by more than one meter by the end of the century. Monitoring and understanding sea level changes requires a combination of different techniques, in order to capture processes that take place in the open ocean as well as those that affect coastal areas. The course is about the observation techniques used to measure sea level variations including their potential and limitations, the analysis of the data provided by the measurement sensors, and their application to the study of the geophysical processes causing sea level variations. The course is structured as follows: Satellite radar altimetry - Basic concept, satellite missions, overview applications, general data processing - From altimeter range to sea surface heights - From sea surface heights to the dynamic topography - Marine gravity and sea floor topography (bathymetry) - Oceanographic applications of satellite radar altimetry Tides and tide gauges - Ocean tides - Tide gauge systems and data analysis - Sea level reconstructions from tide gauge records Sea level change Glacial Isostatic Adjustment Global mean sea level change and regional variability Sea level budgets Past and future sea level change at the Dutch coast.	
Study Goals	After successful completion of this course, the student: - can explain the basic principles of satellite altimetry; - is able to describe the major oceanographic and geodetic applications of satellite altimetry; - is able to process tide gauge records and satellite altimetry data; - is able to analyze past sea-level changes based on tide gauge and satellite altimetry observations; - can explain how different physical processes contribute to sea-level change at global and regional scales.	
Education Method	On a weekly basis: 2 x 2 hours of lecturing 7 hours of self-study and working on graded assignments	
Assessment	The student knowledge is assessed by means of: - graded homework assignments (50% of the final grade); - written examination at the end of the course (50% of the final grade). Judgement: Final grade consists of: two homework assignments (1/4 each), written exam (1/2). Each component needs to be completed with a minimum grade of 5.0 (out of 10.0). During Academic Year 2022/2023, two exams will be offered for students of previous years who have already received a grade of 5.0 or higher for their homework assignments. Other students will only be accepted under exceptional circumstances, to be discussed on an ad hoc basis. Those students will be assessed by means of an oral examination.	
Expected prior Knowledge	Basic use of Matlab (data manipulation and visualisation).	
Academic Skills	Critical thinking. Use of observations to solve practical problems. Awareness of issues related to the use of available observations for the purpose of explaining natural processes.	
Literature & Study Materials	Lecture slides, scientific journal papers	
Judgement	Final grade consists of: two homework assignments (1/4 each), written exam (1/2). Each component needs to be completed with a minimum grade of 5.0 (out of 10.0). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Closed-book exam: only non-graphical calculator allowed.	
Collegerama	No	

CIE4609	Geodesy and natural hazards	4
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course addresses the use of remote sensing, in particular Synthetic Aperture Radar, to monitor or forecast a range of natural hazards, such as earthquakes, volcanic eruptions, or landslides. Particular emphasis will be given to the measurement of surface deformation using satellite radar interferometry (InSAR). Topics to be covered on geodynamic signals include: 1. Introduction to a range of natural hazards (volcanic eruptions, Earthquakes, tsunamis, landslides, land subsidence), link between science and society, mitigation and risk reduction and how geophysical observations (like InSAR) can improve our understanding of these phenomena. 2. Volcanoes: types, dynamics, monitoring (deformation InSAR, GNSS, gas emissions, seismicity, micro-gravity), forecasting, the use of (In)SAR in volcanic studies, linking real data to models. 3. Earthquakes: physical principles, earthquake deformation modeling, probabilistic earthquake forecasting, real-time earthquake prediction, earthquake mitigation. Topics to be covered on geodetic observations are: 1. Basic principles of radar remote sensing, including key performance indicators, such as resolution, sensitivity, spatial coverage and temporal sampling; 2. Fundamentals of SAR image formation, discussing the acquisition geometry, time and frequency domain interpretations, spectral support of SAR images. 3. SAR post-processing (calibration, radiometric corrections, speckle filtering, etc). 4. Interferometric processing steps and time series analysis	
Study Goals	After completion of this course the student will be able to: - Describe the diversity of natural hazards and how science can help society to improve hazard mitigation strategies. - Select which data have to be integrated to understand the dynamics of a volcano and thus improve the forecasting of volcanic eruption - Assess the factors influencing the tsunami risk based on the occurrence of a tectonic Earthquake. - Interpret the performance parameters of a SAR mission, select the most adequate sensor for a given application, and predict the product level performance (e.g. ground deformation uncertainty) - Spectrally analyse SAR data and manipulate the single-look-complex data including sub-look generation, implementation spectral diversity algorithms, and standard multi-looking techniques. - Produce repeat-pass interferograms and geophysically interpret the results. - Derive surface-deformation from interferometric data.	
Education Method	A combination of lectures, supervised lab instruction, self-study and assignments. Some of the assignments use Matlab	
Assessment	Graded assignments and a term paper. Paper: 50% of the final mark Assignments: 50% of the final mark	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIE4611	Geo-measurement processing	5
Responsible Instructor	Dr.ir. A.A. Verhagen	
Instructor	Prof.dr.ing. R. Klees	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	CIE4615 Geoscience and Remote Sensing Fieldwork	
Course Contents	The aim of this course is to convey the necessary knowledge and to develop practical skills for working with random variables, combining measurements for parameter estimation purposes, and testing the validity of measurements and models. Measurements are key to the approach in Geoscience and Remote Sensing exercised in Delft, and support Civil Engineering activities through land surveying. The focus of this course is on the mathematical statistical aspects in the process of going from measurements to results, interpretation, decisions and conclusions. Emphasis is given to develop working knowledge; most of the course effort will be on applications (with actual measurement data). 1. Refresher probability and statistics: random variable, Probability Density Function (PDF), multi-variate distributions, mean, variance, variance matrix, correlation, methods to assess the statistical distribution of empirical data (histogram, Q-Q-plot). 2. Transformation of random variables/vectors, error propagation laws, error budget. 3. Parameter estimation, estimation error, (weighted) least-squares, regression, surface fitting, curve and harmonic function or polynomial fitting, assessment of the fit, geometry of least-squares. 4. Best Linear Unbiased Estimation (BLUE), non-linear least-squares and linearization. 5. Maximum likelihood estimation, estimation of variance factor, confidence intervals and regions, optimality with Gauss Markov model. 6. Detection and validation, probabilities of incorrect decision (level of significance, missed detection, power), statistical hypothesis testing, quality control. 7. Ill-conditioned problems, instability, consequences, constraints, minimum norm solution, regularization.	
Study Goals	After taking this course students should be able to - apply the error propagation laws; - describe and apply the following estimation principles: weighted least-squares, best linear unbiased estimation, maximum likelihood estimation; - set-up an observation model and estimate a set of unknown parameters from a given set of geo-measurements (e.g., remote sensing, GNSS, gravimetry, surveying); - make a quality assessment and interpret the estimation results; - test for errors in measurement data and the observation model; - devise a regularization approach for ill-conditioned problems.	
Education Method	The course is not offered anymore, only the assignments and exam can be taken this year. You can study the course using the online videos, quizzes and materials available via Brightspace. Contact the responsible instructor for more information.	
Computer Use	Python or Matlab can be used for the assignment work. Demos and examples will be provided.	
Assessment	ass. + exam	
Tags	Calculus Lineair Algebra Mathematics Optimalisation Stochastics	
Expected prior Knowledge	General calculus and linear algebra. Basic (BSc) course on Probability and Statistics.	
Academic Skills	Mathematics, critical thinking, analysis, visualisation	
Literature & Study Materials	Hand-outs and videos made available on Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	In case of on-campus exam: Formula sheet (will be provided) and pocket calculator. In case of online exam: all course materials, pocket calculator.	
Collegerama	No	

CIE4615	Geoscience and Remote Sensing fieldwork	3
Responsible Instructor	Dr.ir. A.A. Verhagen	
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	CIE4606, CIE4611	
Course Contents	The aim of this course is to synthesize the knowledge and skills obtained in the first year of the Geoscience and Remote Sensing master track. It includes: 1. Hands-on practice in planning and making observations with different techniques (e.g. GNSS, leveling, total station, gravimeter, meteo stations). 2. Data processing, analysis and interpretation. 3. Background classes on the physical processes being measured. The assignment is given to the students in the form of a project on which they have to work in a team of 8-12 members. The fieldwork will be combined with the one of the AESM402B module.	
Study Goals	The aim of the course is to teach the practical side of making measurements in the field in order to constrain physical processes of the Earth and atmosphere that act at various spatial and temporal scales. After completion of this course the student will be able to: - plan a field campaign that is appropriate for the physical process to be measured, - collect data in the field using different measurement techniques, - explain and quantify the error sources associated with the field measurements, - process and analyse the data collected in the field to give meaningful constraints on the physical process, - effectively communicate with peers, assessors and clients, - contribute to a project as a team player and to the overall project management.	
Education Method	- Instruction in the field - Supervised data collection, processing and analysis - Presentations - Peer and self assessments	
Assessment	review meetings	
Enrolment / Application	An information meeting will be organised in December and May prior to the fieldwork. The course is compulsory for all Geoscience and Remote Sensing students. Information will be available via Brightspace.	
Remarks	The fieldwork will be in the last two weeks of the fourth quarter (exam weeks). Students will have to pay some of the costs for the fieldwork (travelling and stay) themselves. More information will be available on the Brightspace site of this course.	
Tags	Analysis Fieldwork Group Dynamics/Project Organisation Group work Integrated Intensive Involved Modelling Practicals Project Project planning / management	
Expected prior Knowledge	Knowledge related to geo-measurement processing, reference systems, signal analysis and interpretation, Matlab or Python, visualisation tools.	
Academic Skills	Literature study, critical thinking, research skills, discussion with peers and client, presentation, project management, fieldwork	
Literature & Study Materials	All materials will be provided on site.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

CIE4616	Remote Sensing and Big Data	5
Responsible Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Many image processing and analysis techniques have been developed to aid the interpretation of a range of images (e.g. satellite remote sensing, but also other types of raster datasets like climate data) and to extract as much information as possible from the image data. Recent advances in remote sensing and computer science has moreover resulted in an explosive growth of image data sets and data analysis techniques such as machine learning. This evolution is both a challenge and opportunity as it requires specific techniques to explore, analyse, and leverage the data. This course provides an overview of tools and techniques to explore, analyse, and visualise the image data and on the implementation of big data analytics techniques on these data to exploit the growing data archives.	
Study Goals	At the end of this course you will be able to: - Explain the basics of image processing and big data analytics for (remote sensing) image processing - Apply a variety of more advanced image processing techniques (pixel-based, object-based) to extract information from (remote sensing) imagery - Identify a variety of remote sensing problems where image processing can help to translate data into information - Implement big data analytics techniques to implement the image processing techniques on big data archives (e.g. satellite data sets). - Design and code a processing chain for implementation of the image processing techniques and big data algorithms	
Education Method	The course will be given this year in different format where students will have access to the online lecture recordings of 2021-2022 (i.e. self study) in combination with a weekly (online) contact moment where students can discuss/reflect on the recordings. The content of the course will be a combination of recorded introduction videos (50%) and hands-on assignments (50%) where students will implement the learned techniques on a variety of (big-) (remote sensing) image data sets. The hands-on part will be a combination of assignments (where each group/students gets exact the similar assignments) and project work (where each where each group/students does a more in-depth analysis of one specific topic). During the assignments, students will learn to apply scalable big data algorithms on different platforms (e.g. Google Earth Engine, Google Cloud, Amazon Web Service, etc.)	
Assessment	The final score will consist of : - 50% assignments - 50% project work To pass the course, students need a 6 for the whole course and minimum a 5 for both the assignment and project work.	
Expected prior Knowledge	Generic programming skills.	
Academic Skills	Geo-data analysis, visualisation and programming.	
Literature & Study Materials	Material will be posted on Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No exam, but assessment based assignments and project work.	
Collegerama	No	

CIE5401	GIS & Remote Sensing for Water Resources	3
Responsible Instructor	Dr. S.C. Steele-Dunne	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course discusses Geographical Information Systems (GIS) and remote sensing in water resources. The course is designed to allow students to gain hands-on experience by applying GIS (QGIS and GRASS) and remote sensing data for different applications in water resources. In addition to the common GIS tasks of visualizing and analyzing data, we will use GIS tools to delineate watersheds, and to perform land classification and surface water identification and data analysis. The remote sensing component is focused around measurements in the visible, thermal and microwave regions of the spectrum. We will introduce the physics behind the measurements, and illustrate how observations in each region yield invaluable data for water resources. Remote sensing data will be used to analyze elevation and land cover distribution, as well as precipitation, soil moisture and evaporation patterns in a basin. In exercises and lectures, different tools and applications are offered. For each application, assignments are given to allow students to acquire relevant skills. The course structure combines assignments and introductory lectures. Each week participants work on one assignment. These assignments are discussed in the next lecture and graded. Each week a new assignment is introduced, together with supporting materials (an article discussing the relevant application) and lectures (introducing theoretical issues). The study material of the course consists of a study guide, assignments, lecture material and articles. The final mark is the average of the grades of the individual assignments.	
Study Goals	After this course a student can: 1) Explain what a GIS is, and to describe its key features 2) Use GIS to visualize and analyze data 3) Use GIS tools to work on real WRM problems 4) Understand important terminology in remote sensing 5) Explain the advantages and disadvantages of measurements in different parts of the EM spectrum 6) Use real remote sensing data to study problems in WRM	
Education Method	Lectures, exercises	
Literature and Study Materials	Available through Brightspace	
Assessment	Assignment	
Expected prior Knowledge		
Academic Skills		
Literature & Study Materials		
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam		
Collegerama	No	

CIE5817	Assessment of Transport Infrastructure and Systems	4
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	N. Mouter	
Contact Hours / Week x/x/x/x	This course is discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	**** discontinued. Exam opportunities are still offered for in the academic year 22/23. See for details under assessment and judgement. For more information on support material and possible contact hours: please check the Brightspace page of this course. **** The course is structured in four main, content related lecture topics, and an assignment: Methods: The basic principles of qualitative, quantitative and monetary assessment methods will be introduced of which three more in-depth. Impacts: Assessment methods compare alternative proposed solutions for transport problems in a systematic and 'fair' way. A first challenge is to determine what type of effects (i.e. impacts) to take into account and how to gather - preferably quantitative - data on these effects. For a monetary analysis, a second important step is to attach a monetary value to these effects. Application: The application of (s)CBA and/or MC(D)A methods for traffic management, public transport, information provision and automated transport projects will be discussed as examples. In policy practice: The use of quantitative evaluation methods in actual policy processes is discussed and analysed. Assignment: In the assignment the theoretical approaches introduced during the lectures will be put in practice. The assignment includes different cases, different methods, and a sensitivity analysis. The course introduces, discusses and challenges to apply formal qualitative and quantitative assessment methods for transport infrastructure and systems projects. The lectures will provide insights in methods and approaches of formal qualitative and quantitative evaluation methods that are used to compare alternative solutions for transport problems. The course will help to be able to understand the approaches as used in private and public practices and to interpret the results. Introduction on qualitative and quantitative evaluation methods: Financial analysis: private sector investors view on investments, cash flows, time value of money, internal rate of return, and project risk assessment. (Societal) Cost Benefit Analysis: basic concepts and principles, monetisation of cost and benefits, social discount rate, net present value estimation. Multi-Criteria (Decision) Analysis: basic concepts and principles, the role of criteria in decision-making, criteria weighting, sensitivity analysis. Alternative assessment approaches, specifically for (a) taking into account a broader perspective of sustainability and welfare and (b) methods for risk assessment. Assessing impacts, introduction on most relevant criteria/factors: Travel cost impacts assessment; (Traffic) Safety impacts assessment; Environmental and ecological impacts; Land use impacts; Social impacts (incl. Distribution Effects). Application of assessment methods: specific cases, policy practice: Transport infrastructure projects (e.g. road infrastructures), Transport systems (e.g. public transport systems) and intelligent transport systems, and automated driving), Dealing with uncertainties (creating alternatives, scenario analysis, sensitivity analysis), Usage and usability of assessments private and public planning process, in different national and cultural settings and against the background of societal challenges. Study load: 42 hours lectures, 14 hours preparation lectures, 21 hours assignment, 35 hours exam preparation	
Study Goals	The course aims to enable students to understand and apply qualitative and quantitative assessment methods for large scale transport projects, to critically analyse and evaluate their use and to design a basic assessment model. By the end of the course students are able to: LG1: Identify and describe the key impacts/factors to be included into qualitative and quantitative assessments. Identify concepts and main methods for quantification and monetary valuation. LG2: Understand evaluation methods for non-market costs and explain the concept of time value of money, the related present value principle and the use of discount rates; LG 3: Apply financial analysis method, cost-benefit analysis and multi criteria analysis for basic cases, including finding and estimating appropriate values for criteria and weighing factors; LG 4: Analyse and evaluate the use of qualitative and quantitative assessment approaches, taking into account objectives of stakeholders, possible cultural differences in value systems and possible biases in the various steps of the analysis process; LG5: Create a cost-benefit analysis model and a multi criteria analysis model (spreadsheet); estimate appropriate values for criteria for future transport concepts.	
Education Method	The course consists of lectures and an assignment. The lectures focus on giving introductions and background information on evaluation methods, and examples of cases and applications thereof. Lectures are designed in a way to spark discussion and to interact. The assignment is performed by student groups. In the assignment students apply knowledge derived from lectures and additional information gathered by themselves. The assignments require performing spreadsheet modelling/calculations and reporting.	
Course Relations	CIE4801-18 Transport Modelling CIE4831-18 Empirical analysis for Transport & Planning	
Prerequisites	Micro-economics (costs factors, benefits, monetarisation), Transport planning and engineering (main characteristics of transport modalities, the notion of reliability in travel time, intermodal travel, transport networks and links), Transport modelling (i.e. notions of 'value of time', modal split/mode choice, route choice, network and link characteristics).	
Assessment	Group assignments and written exam (assignments is group work, there will be no possibility to do an assignment in academic	

Tags	year 2022-2023) Economics Group work Policy Analysis Transport & Logistics
Expected prior Knowledge	Micro-economics (costs factors, benefits, monetarisation), Transport planning and engineering (main characteristics of transport modalities, the notion of reliability in travel time, intermodal travel, transport networks and links), Transport modelling (i.e. notions of 'value of time', modal split/mode choice, route choice, network and link characteristics).
Academic Skills	Students are expected to exercise the following academic skills: Thinking (critical, analytical) Interpretation Cooperation Reasoning / arguing Judgemental skills Concise reporting
Literature & Study Materials	Lecture notes, slides and references to literature (books, articles available online via TU Delft Library account), example exam questions.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Closed book exam. (Non-programmable) calculator.
Collegerama	No

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. C.M. Jonker	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	Written exam and practical assignments. The grade of the course is determined by the grade of the written exam (80%) and the mean grade of the practical assignments (20%). There are no resit possibilities for the practical assignments. There will be a resit possibility for the written exam.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l_0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l_0 norm is shown to be NP-hard, and several classical approximate algorithms like l_1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Books	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%).	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovy	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers.	
Books	Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: - A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

IN4252	Web Science & Engineering	5
Responsible Instructor	J. Yang	
Responsible Instructor	Dr. C. Lofi	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	The student learns the important principles and concepts of Web-based Information Systems, and understands the main research challenges in the area. The student has knowledge about the main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. The student has knowledge of the main principles and techniques for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modelling. The student learns the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science. The student is able to write a paper contributing to Web Science based on a problem in the field of web-based information systems.	
Education Method	The education includes: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper and contribute to relevant research. Lectures will be not each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessment happens on the basis of the small assignments (accompanying the lectures), for 20% of the grade, and the large assignment (writing the web science paper), for 80% of the grade. All assignments must be completed by the indicated deadlines. Students must get a pass for each assignment. The final paper needs to be graded with a 6.0 or higher.	
Special Information	disclaimer: information may change depending on the developments around the coronavirus. Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters. The course is completed with a final paper; students have one chance per year to re-submit a paper if it is judged insufficient.	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: - o identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams - o responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles - o adapting with flexibility to the dynamic requirements of a complex external assignment - o translating feedback received into proactive personal development steps - o formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: - o identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints - o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	Project assessment will be based on a combination of: - (~50%) product grade: unique for the whole group, based on both the game itself and the required documentation; - (~45%) process grade (individual), including personal contribution, performance, attitude, and peer evaluation; - (5%) final presentation. The commissioner will be involved both as advisor and as assessor. The final documentation will include writing a scientific paper and actually submitting it to a conference on serious games and/or their application. The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5).	
Co-Instructor	Dr. R. Marroquim	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strengths and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	Weekly individual assignment, weighting 10% of the final grade. Two group projects for IR and NLP, each weighting 45% of the final grade. The group assignment is performed collectively, but graded individually. Assignments have no resit opportunities. disclaimer: information may change depending on the developments around the coronavirus.	

RO47003	Robot Software Practicals	5
Responsible Instructor	J.F.P. Kooij	
Instructor	G.A. van der Hoorn	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming experience (e.g. python or matlab) at the BSc level, familiarity with the use of functions, variables, and control structures (for-loops, if-statements, etc.).	
Course Contents	This course teaches students the basics for creating and collaborating on academic robotics software. It aims to give students sufficient knowledge and experience such that they can proceed learning more about robotics programming by themselves as needed. The course does this by providing an overview of key technologies, and letting students develop basic skills often used in real world robotic software development. The main objective is to obtain hands-on experience - beginners level - with Linux, Git & Gitlab, C++, ROS, and integrate with common external libraries such as OpenCV and PCL. This course does not teach new robotics theory (e.g. perception, planning, control, etc.) which is covered in other courses. The allotted 5 ECTS means that students are expected to invest approximately 11 hours/week outside of contact hours. Be aware that active participation in all the lab and peer-review assignments is required, hence it is not recommended to do this course if you cannot commit to a regular time investment over the whole education period.	
Course Contents Continuation	Linux: Although Windows is the most prevalent operating system on desktop PCs, Linux became very popular for embedded systems such as robots. Developing for embedded Linux systems is most easily done in Linux itself, and this course aims to familiarize students with the use of Linux on the desktop and terminal. It includes the following topics: OS architecture, File system, Shell, Scripting. Git/gitlab: All exercises will be submitted using Git, a widely used version control system to manage and collaborate on coding projects. During the course, students will practice basic tasks with git, such as solving code conflicts, and merging the work of lab partners. Using an issue tracker and giving constructive feedback to peers is also an important aspect that will be practiced. C++: C++ is one of the most relevant programming languages for robotics. Being an object oriented language brings great advantages compared to its predecessor C. This practical gives students practical knowledge on C++ programming. The course encompasses the following topics: Introduction to C++, common compilation tools, basic programming in C++, classes and objects. ROS: The Robot Operating System (ROS) is a flexible framework for writing robot software. It is a collection of tools, libraries, and conventions that aim to simplify the task of creating complex and robust robot behaviour across a wide variety of robotic platforms. During the course, students will learn to use existing ROS tools, create their own software components in C++, integrate them into an application, and test their solutions using a physics simulator. others: You will also get some practice using external libraries in your robotics project, such as OpenCV or PCL. OpenCV is a computer vision library designed for computational efficiency and with a strong focus on real-time applications. The Point Cloud Library (PCL) is an open-source project for point cloud processing. Note that this course is <i>*not*</i> about understanding the methods or theory on which these libraries are based, but rather on integrating the provided functionality in novel software components.	
Study Goals	----- Knowledge, insight, judgment and skills ----- After completing this course, students will be able to 1. use the command line on a Linux system, perform basic file operations and job management tasks, and automate tasks with shell scripts; 2. use Git to commit and integrate incremental code changes together with a lab partner, and resolve any code integration conflicts; 3. use a Gitlab server to exchange code changes, and use its issue tracker to provide and receive feedback on each other's code; 4. solve programming tasks in C++, compile the code using Make, debug compile-time, link-time and run-time problems; 5. create new C++ ROS packages, use common ROS development tools, and utilize the ROS API in their C++ programs; 6. investigate and integrate external libraries into their own ROS components, such as OpenCV or PCL; 7. add code comments and basic documentation to their code, find and interpret documentation of external libraries; ----- Transferable and interpersonal skills ----- 8. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback; 9. collaborate and coordinate with lab partner on coding tasks.	
Education Method	Lectures, which will introduce new course content (2 hours per week). Interactive Sessions will provide class wide feedback on past lab assignments, tutorials, quiz exercises, Q&A (2 hours per week) Practicum session, during which there will be TAs around to help you understand the lab assignments (2 hours per week) Robotics Software Practicals focuses on improving programming and collaborative software development skills. The main method to develop these skills is by gaining hands-on experience through lab assignments, and receiving feedback on the submitted work. Lectures will introduce the various topics, provide context and tips, and will discuss common pitfalls and solutions. Still, the nature of the assignments means that students are expected to invest sufficient time for self-study and practice outside the fixed lab and lecture contact hours. Lab assignment There are 4 obligatory lab assignments, each covering two weeks, namely 1. Linux, Git (Formative) 2. C++ programming (Formative) 3. ROS (Formative) 4. Final Assignment (Summative, demonstrates capacity to use Linux, Git, C++, and ROS). A manual will be provided for each lab assignment before the start of the first practicum session for the assignment. The manual will contain detailed instructions, guidelines, optional exercises, and an assignment that must be submitted to an online repository for your lab group.	
Literature and Study Materials	1. A Gentle Introduction to ROS, Jason M. OKane. http://www.cse.sc.edu/~jokane/agitr/ 2. Provided lab manuals, and lecture slides that will be made available on Brightspace	

Practical Guide

For the practicum assignments, bring a reasonably recent laptop with a x86-64 CPU (i.e. a normal Intel or AMD CPU; but NOT the new MacBook Air with M2 chip, NOT a Chromebook, etc. These use different CPU architecture which means that the software we provide will not run! A MacBook with an Intel x86 CPU is fine). A laptop with an Nvidia GPU would make the robot simulations run even smoother.

Please note the following lab rules:

The first three lab assignments are made in groups of two, as you will learn to collaborate on code with your lab partner using gitlab. You can find a lab partner using the special Brightspace forum, but of course also by talking to others after the first lecture, or by looking for a Brightspace lab group with a single enrolled person.

To enrol in the practicum, and therefore participate in the course, you and your partner *must* enrol on Brightspace in a lab group before the lab enrol deadline (this deadline will be announced in first lecture).

If you plan to drop the course, inform the lab coordinator in time before the start of the next lab assignment, but while you are still enrolled, you should finish the ongoing lab assignment with your lab partner.

The scheduled practicum sessions are intended for you to obtain help on the assignments from the TAs. Outside the practicum sessions, you can post questions on the TA forum on Brightspace.

To submit your work, you and your partner must push your contributions to a provided gitlab repository. Note that using git and the gitlab webserver for code collaboration is part of the learning objectives of this course. You are responsible for testing the code that you hand-in via gitlab, so make sure that you have tested your work thoroughly, and that the correct version has been uploaded to your gitlab repository.

For the first three lab assignments, there will be a peer-review task after the lab deadline has passed. You will learn during the course how to provide feedback, and how to use the gitlab issue tracker to manage issues and comments. The lectures will afterwards discuss positively peer-reviewed lab solutions with all students for additional feedback by a teacher.

Active participation in each lab assignment is required to participate in the next assignment. Active participation entails that you submit your solution before the given deadline, and that you complete the required peer-review tasks.

You may only use the gitlab code repository that the lab coordinator will create for your group. You are *NOT* allowed to put your coding homework on github.com, gitlab.com, bitbucket.com, google drive, dropbox, etc. (neither with public nor private settings!)

TU Delft ethical guidelines apply to reports, code, etc. If your name is on it, you claim it is your work, do *not* copy parts of other student groups code. You are *not* allowed to share your code, solutions with others apart from your lab partner and TAs. Generally, do *not* show your groups work to other students. Plagiarism will be reported to exam committee.

Assessment

Assessment overview

Formative: First three lab assignments, peer-review tasks, discussion on common mistakes and solutions of past hand-in assignments during the lecture by teacher, multiple-choice practice questions.

Summative: The final lab assignment, and digital exam.

Grade calculation

You final course grade (minimum is a 6 to pass) determined by

Final lab assignment (grade, weight 25%, minimum a 5.5 to pass),

Digital exam (grade, weight 75%, minimum a 5.5 to pass)

Any penalties (see Deadlines below)

A resit will be available for the exam, but not the final lab assignment.

Lab assignments

You must enroll for the obligatory lab assignments before the announced enrollment deadline (given during first lecture), so that the lab coordinator can fix the groups and setup everybody's code repositories in time. If you do not enroll, you cannot make the lab assignments, or make the exam.

In the first three assignments you must collaborate with a lab partner. Your group will receive and give formative feedback through student peer-review (what goes well, what to improve).

The fourth and final lab assignment is made individually, and you receive a grade (weights 25% for final grade), based on a rubric for the quality of the solution, code, documentation, and use of Git. This assignment requires you to apply all knowledge from the previous lab assignments to solve a well-defined robotics programming task.

Your (groups) lab work is evaluated on the following three Lab Criteria (LC):

- LC1: solution quality (does the solution solve the given task?)
- LC2: code quality (are there any bugs or problems with the code?), and
- LC3: documentation quality (is there a clear readme, does the code contain sensible comments?).

Deadlines, submission requirements, and penalties

While most of the lab assignments are formative, your active participation is required and there are strict submission and peer-review requirements:

- LC4: Each partner must fulfil each lab's minimum submission requirements regarding use of Git, and Gitlab, file layout, and build instructions, before the lab deadline. Each assignment's unique submission requirements will be listed in the corresponding lab manual.
- LC5: You must participate in the peer-review tasks for the first three lab assignments, and complete these before the given peer-review deadlines.

Failing to comply with these criteria will result in penalties on your final *course* grade:

- After lab submission deadline: -1 grade point per day for not completing the assignments' minimum submission requirements, with a maximum of -2 per assignment. Not enrolling in time before the lab automatically results in -2.
- After peer review deadline: immediate -0.5 grade point for not fully completing the peer-review task (only applies to those enrolled for the lab).

NB:

- These criteria apply to both lab partners individually. If you comply with the criteria but your partner does not, only your partner will receive the penalty.
- Penalties apply to your final course grade, so if you do not submit for one lab (-2 penalty) you could only complete the course with an 8 at best.
- There are NO redos or extra assignment opportunities to remove a given penalty. You will have to redo the lab next year and comply with the deadlines then for a clean slate.

Exam

There will be an individually-made digital exam, for which you will receive a grade (weights 75% for final grade, a minimum of 5.5 to pass). Questions are based on content discussed in the lectures, and on practical situations that you have encountered during the lab assignments. The exam will also require you to reflect on and refer to your own coding solutions for the final lab assignment, therefore to participate in the exam you *must* have participated in the final lab assignment. Practice questions will be provided before the exam so you can assess your own understanding first.

Department

3mE Department Cognitive Robotics

SEN9235	Game Design Project	5
Module Manager	Dr.ir. G. Bekebrede	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Dr.ir. G. Bekebrede	
Instructor	Dr. L.J. Kortmann	
Instructor	Dr. M. Freese	
Responsible for assignments	Dr.ir. G. Bekebrede	
Co-responsible for assignments	Dr. L.J. Kortmann	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	The project is mandatory for the Advanced Modeling, Gaming and Design specialization) of COSEM, but open to students from all other faculties and outside the university.	
Expected prior knowledge	Bachelor degree	
Summary	The simulation game design project offers conceptual insights as well as hands on experience with simulation-gaming (SG). SG is an established field of practice with proven value for instance in the field of spatial and urban planning, ecology, engineering and design, public administration, business management, learning, research and consultancy. The staggering growth and success of the video gaming industry has triggered the interest in simulation games from paper based games to video games for learning and policy making - even more. Besides creativity and enthusiasm, there are no entry requirements. In this course, students acquire advanced knowledge in, and experience with, gaming-simulation for serious use, i.e. for policy- and decision support, organization and management and professional learning.	
Course Contents	Experienced speakers, from consultancy organizations (s.a. Accenture), game designers and universities, will give lectures on history, future, design, typology and facilitation of SG. (We will discuss the role of computers and video games, but emphasis is NOT on technology!) We will have ample opportunity to try out a number of (role-playing, board etc.) games that hold powerful messages about decision-making and management. Most of the work will be done in small teams, in which students will develop their own (non-digital) game to support a real-life or mock up case of policy/decision-making or organization and management. In a small group, you will also work on a scientific question related to game design.	
Study Goals	After completion of this course the student will have acquired knowledge and insights about: The history, backgrounds, key-concepts, formats and applications of simulation games. After completion of this course the student will be able to: 1. Design a (prototype) of a simulation game to be used for learning, research or intervention 2. Apply a game design cycle (or systematic design steps) 3. Define, conceptualize and construct the various game components 4. Facilitate simple simulation games	
Education Method	(Guest) lectures, workshops, game-play, game design, writing an essay Intensive workshops and group meetings Student group work	
Computer Use	Not likely; dependent on selected case. Possibility playing a computer game	
Course Relations	SEN9110 Simulation Masterclass (5 EC) Advanced Agent Based Modeling (5 ECTS) in4302 Building Serious Games (5 EC)	
Literature and Study Materials	To be announced (books and articles on gaming/simulation and game design). Reading materials will be made available via Brightspace.	
Assessment	Attendance and active participation during classes and workshops. Evaluation of game design and project result. Quality of the essay and active participation in the discussion	
Enrolment / Application	Registration through Brightspace enrollment	
Special Information	Geertje Bekebrede (g.bekebrede@tudelft.nl)	
Remarks	Attendance during the class meetings is required.	
Elective	Yes	
Tags	Challenging Design Group work Intensive Project	
Targetgroup	Master SEPAM students of the modelling, simulation and gaming specialization(MSG) Elective course for students SEPAM, EPA, MOT, other faculties such as Architecture, Industrial Design, EWI, Civil Engineering etc.	
Category	MSc level	

WI4227-14	Discrete Optimisation	6
Responsible Instructor	Dr. F.M. de Oliveira Filho	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge: * Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: * Proficiency in reading and writing proofs * Sufficient mathematical maturity	
Course Contents	Discrete optimization is about the problem of finding a best solution among a (finite) set of feasible solutions. A well-known example is the traveling salesman problem, where we are asked to find a shortest tour in a graph visiting every node exactly once. This course is an introduction to the area, focusing on fundamental methods like linear programming and its theoretical framework (polyhedra, integrality, etc.). The following topics will be covered: - Convex sets and their structure. - Polyhedra, cones, and their structure. - Linear programming, the simplex method. - Systems of linear inequalities, Fourier-Motzkin elimination. - Farkas's lemma, linear programming duality. - Several applications and examples: flows in networks, matrix games, bipartite matching, etc.	
Study Goals	By the end of this course, students will be able to - Recognize convex sets and their applications; - Apply the theory of convexity to concrete examples; - State the main results/theorems concerning polyhedra; - Model problems as linear programming problems; - Recognize total unimodularity and use it to prove integrality of polyhedra; - Construct the dual of a linear program.	
Education Method	Lectures	
Literature and Study Materials	The course is based on lecture notes and no specific book will be followed.	
Books	No specific book will be followed in this course. The following books contain most of the course's material: * A. Barvinok, A Course in Convexity, American Mathematical Society, Providence, 2002. * V. Chvatal, Linear Programming, W.H. Freeman and Company, New York, 1983. * W.J. Cook, W.H. Cunningham, W.R. Pulleyblank, and A. Schrijver, Combinatorial Optimization, John Wiley & Sons, New York, 1998. * A. Schrijver, Theory of Linear and Integer Programming, John Wiley & Sons, New York, 1986.	
Prerequisites	* Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: Proficiency in reading and writing proofs Sufficient mathematical maturity	
Assessment	Up to 20% of the final grade will be determined by exercise sheets, the rest will be determined by a final exam.	
Permitted Materials during Tests	No materials are permitted during the test (except pencil, ruler, eraser etc). In particular, calculators are not allowed.	

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